Efficient sequential learning and counterfactual policy learning

Houssam Zenati

Inria Saclay (Mind)

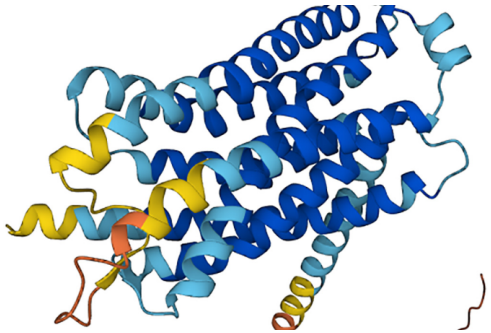
March 12, 2024

Mind-Soda group meeting

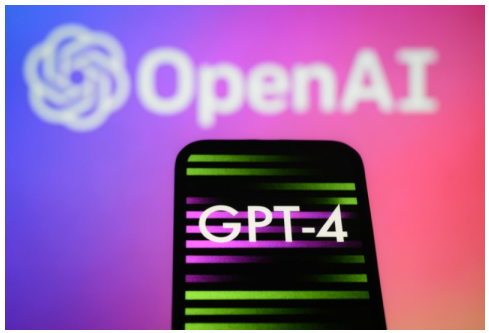
Successes of artificial intelligence



AlphaGo (2016)



AlphaFold (2018)



ChatGPT (2022)

We often use machine learning/ artificial intelligence to make **decisions**, not only **predictions**.

News feed

Online Retailing

Entertainment

Autonomous driving

Decision making systems

In interactive systems, **full information** is not always available. An agent learns to take "better" **actions** so as to optimize a **feedback** signal...



Context x (user info) **Policy** π selects **action** a **Feedback** y

This setting is ubiquitous in many applications.

Decision making systems

In interactive systems, **full information** is not always available. An agent learns to take "better" **actions** so as to optimize a **feedback** signal...



Context x (user info) **Policy** π selects **action** a **Feedback** y

This setting is ubiquitous in many applications.

- online bidding, ad allocation optimization (Criteo, Google Ads)

Decision making systems

In interactive systems, **full information** is not always available. An agent learns to take "better" **actions** so as to optimize a **feedback** signal...



Context x (user info) **Policy** π selects **action** a **Feedback** y

This setting is ubiquitous in many applications.

- online bidding, ad allocation optimization (Criteo, Google Ads)
- content personalization (Netflix)

Decision making systems

In interactive systems, **full information** is not always available. An agent learns to take "better" **actions** so as to optimize a **feedback** signal...



Context x (user info) **Policy** π selects **action** a **Feedback** y

This setting is ubiquitous in many applications.

- online bidding, ad allocation optimization (Criteo, Google Ads)
- content personalization (Netflix)
- music/playlist/podcast recommendation (Spotify, Deezer)

Decision making systems

In interactive systems, **full information** is not always available. An agent learns to take "better" **actions** so as to optimize a **feedback** signal...



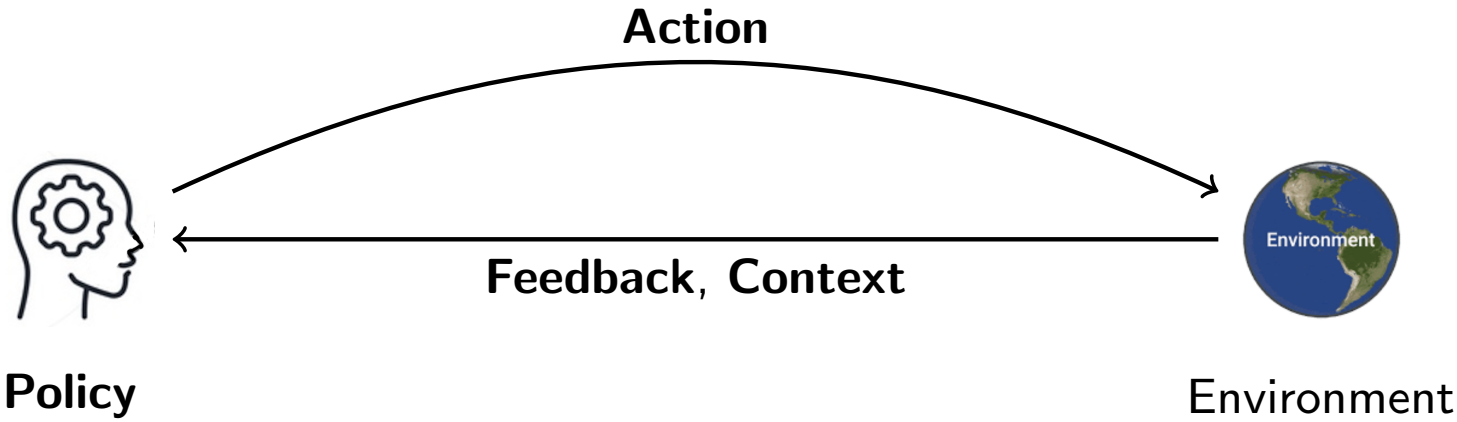
Context x (user info) **Policy** π selects **action** a **Feedback** y

This setting is ubiquitous in many applications.

- online bidding, ad allocation optimization (Criteo, Google Ads)
- content personalization (Netflix)
- music/playlist/podcast recommendation (Spotify, Deezer)
- precision medicine (Pharma Techs)

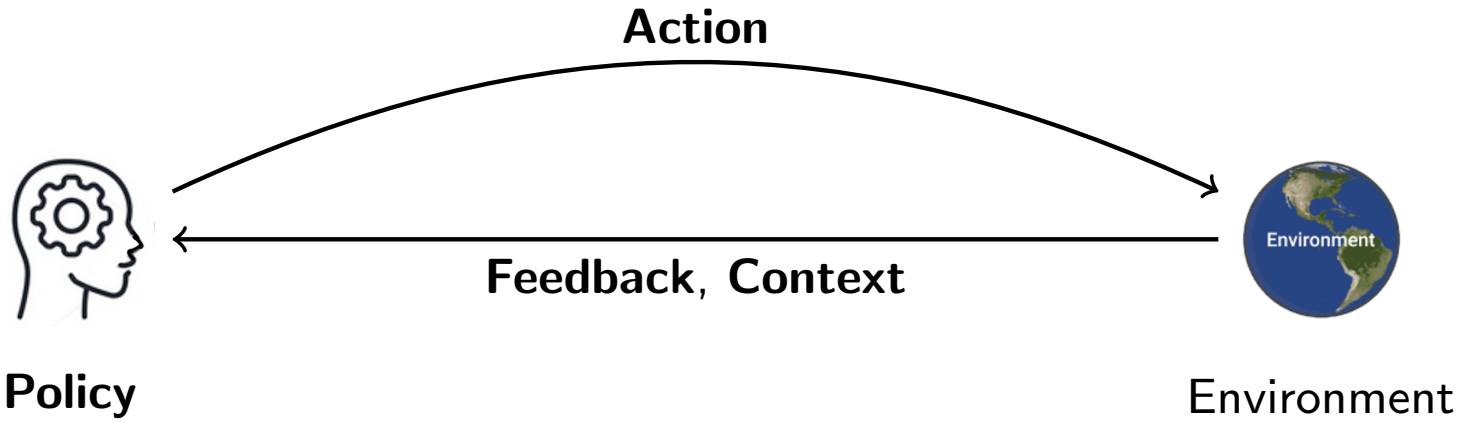
Offline versus online Learning

Online agents learn by **interacting directly** with the environment.

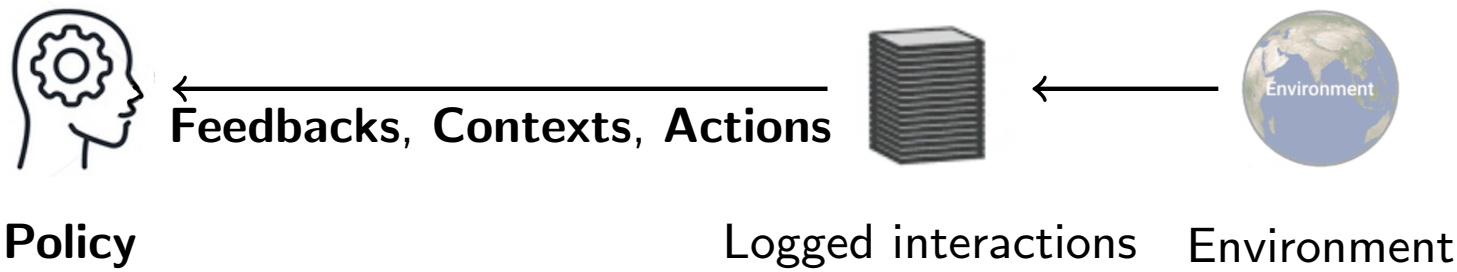


Offline versus online Learning

Online agents learn by **interacting directly** with the environment.

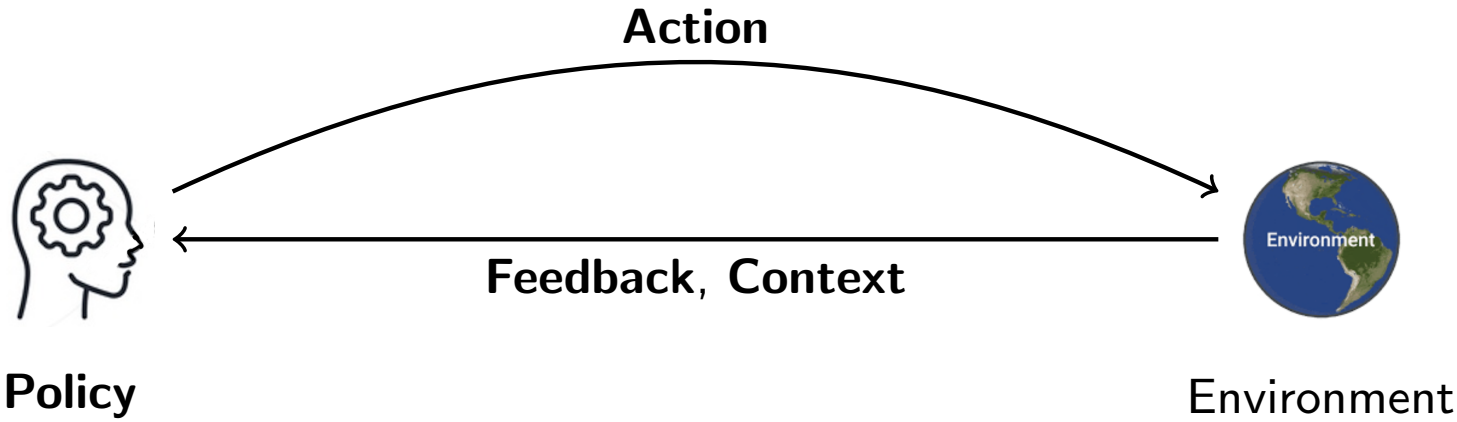


Offline learning methods **leverage past** "logged" **interactions**.

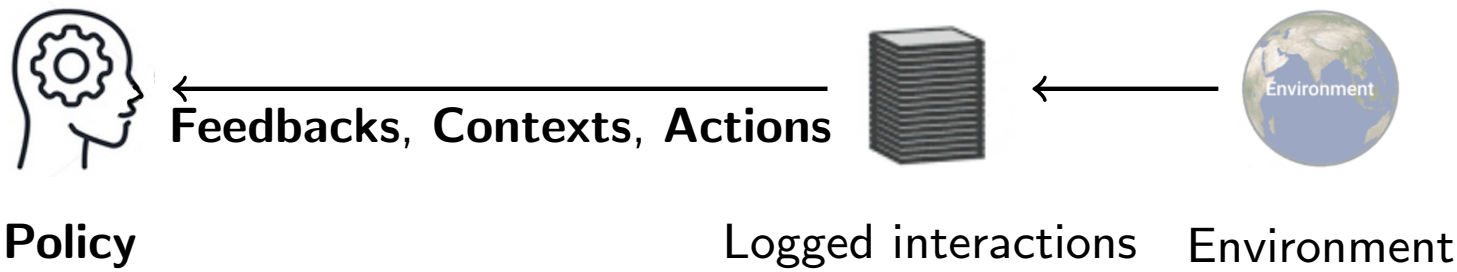


Offline versus online Learning

Online agents learn by **interacting directly** with the environment.



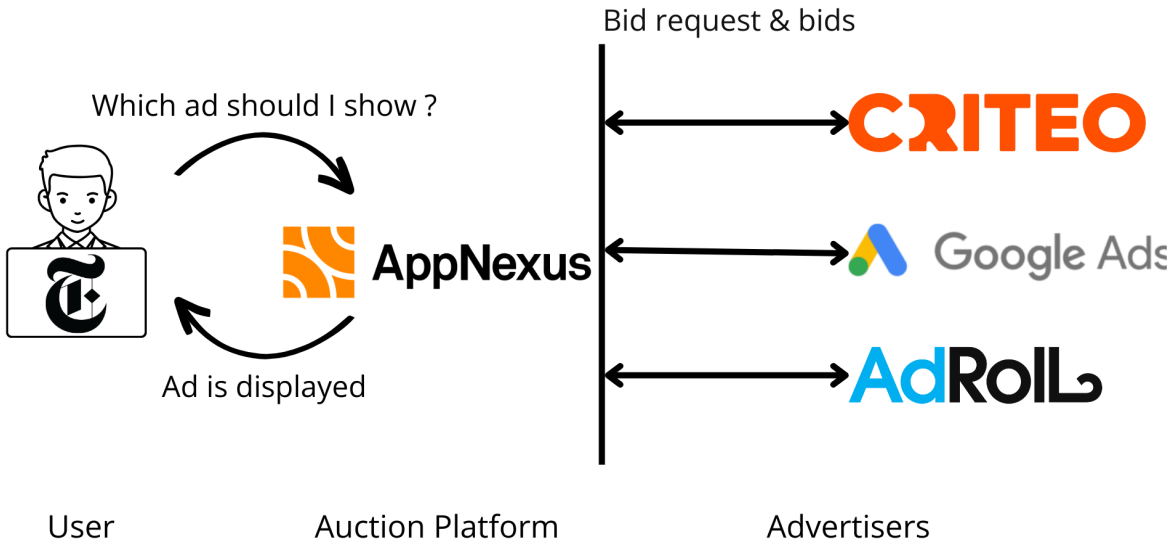
Offline learning methods **leverage past** "logged" **interactions**.



In real-life applications such as advertising, personalized medicine, offline learning may be preferred because of **exploration** or **safety** constraints.

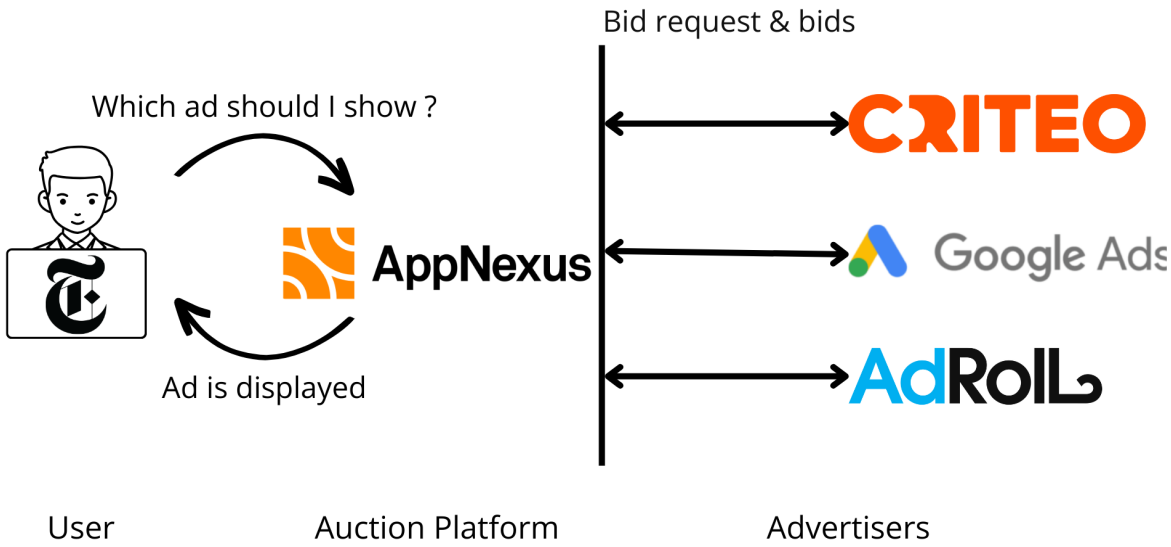
Example 1: Online bidding

The online bidding task involves a user, a publisher, an auction platform and advertisers.



Example 1: Online bidding

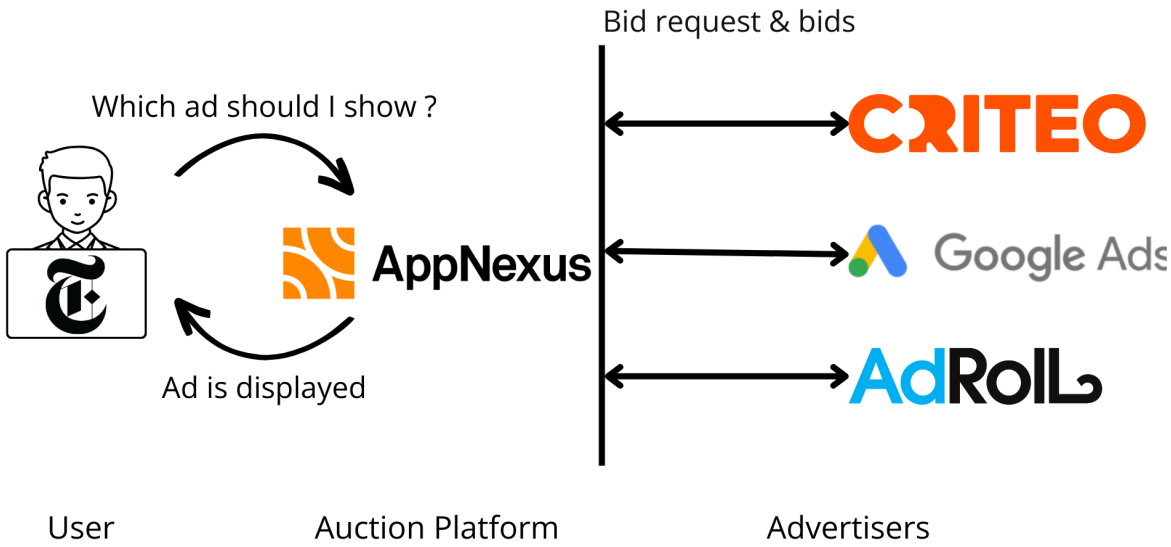
The online bidding task involves a user, a publisher, an auction platform and advertisers.



- **context**: user info, website context, auction details

Example 1: Online bidding

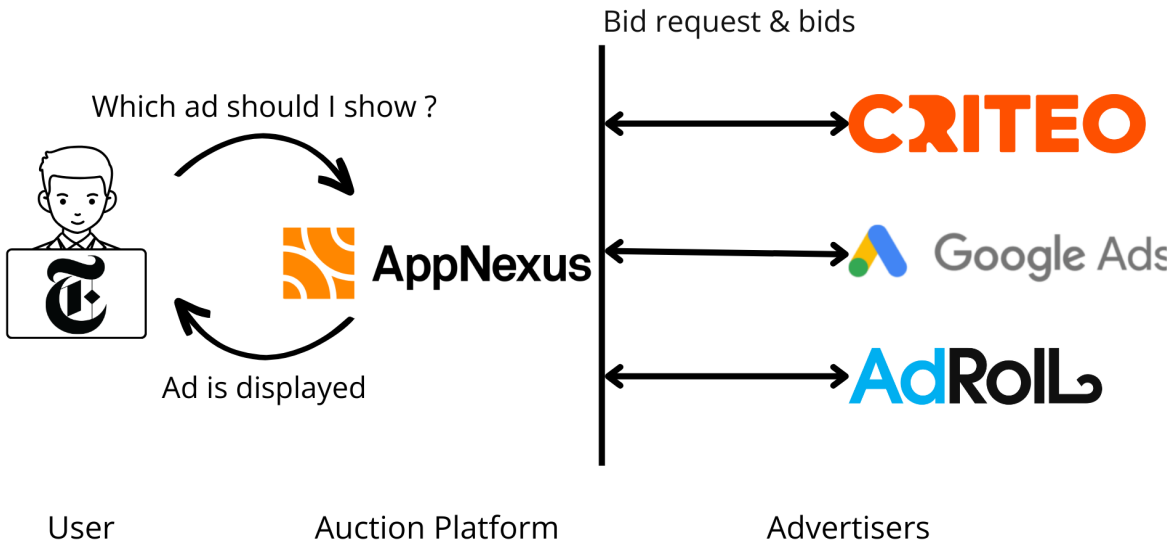
The online bidding task involves a user, a publisher, an auction platform and advertisers.



- **context**: user info, website context, auction details
- **action**: bid value

Example 1: Online bidding

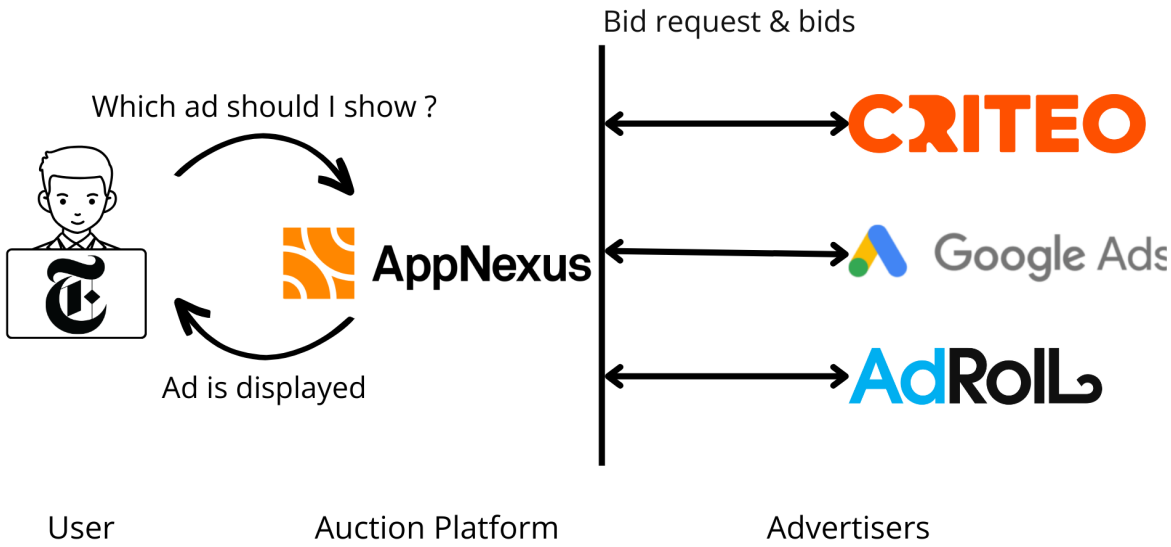
The online bidding task involves a user, a publisher, an auction platform and advertisers.



- **context**: user info, website context, auction details
- **action**: bid value
- **feedback**: click, revenue, cost, etc..

Example 1: Online bidding

The online bidding task involves a user, a publisher, an auction platform and advertisers.



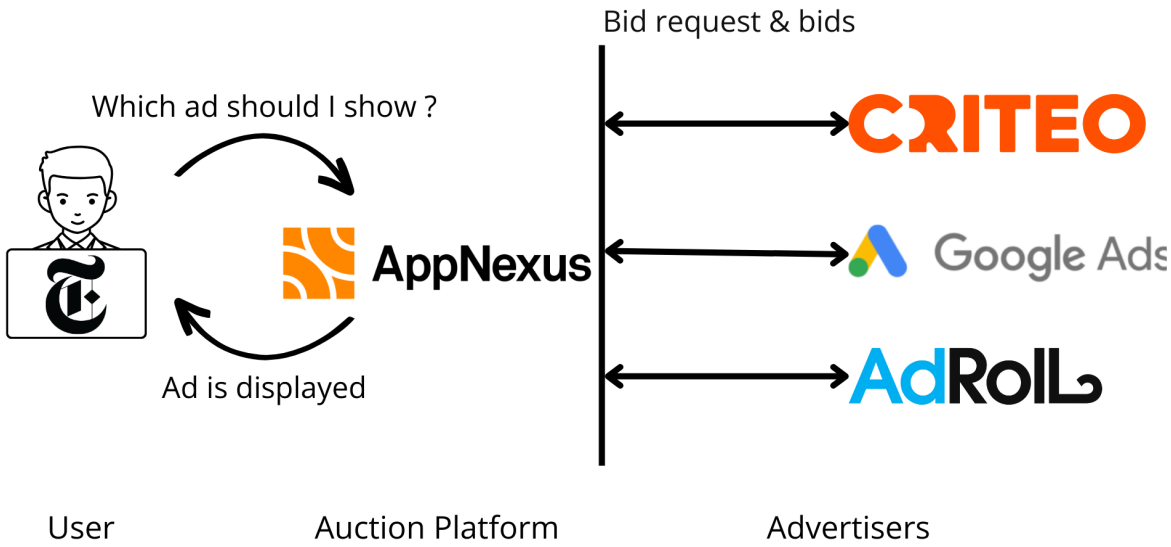
- **context**: user info, website context, auction details
- **action**: bid value
- **feedback**: click, revenue, cost, etc..

Bids have continuous values:

- Need for **continuous**-action methods

Example 1: Online bidding

The online bidding task involves a user, a publisher, an auction platform and advertisers.



- **context**: user info, website context, auction details
- **action**: bid value
- **feedback**: click, revenue, cost, etc..

Bids have continuous values:

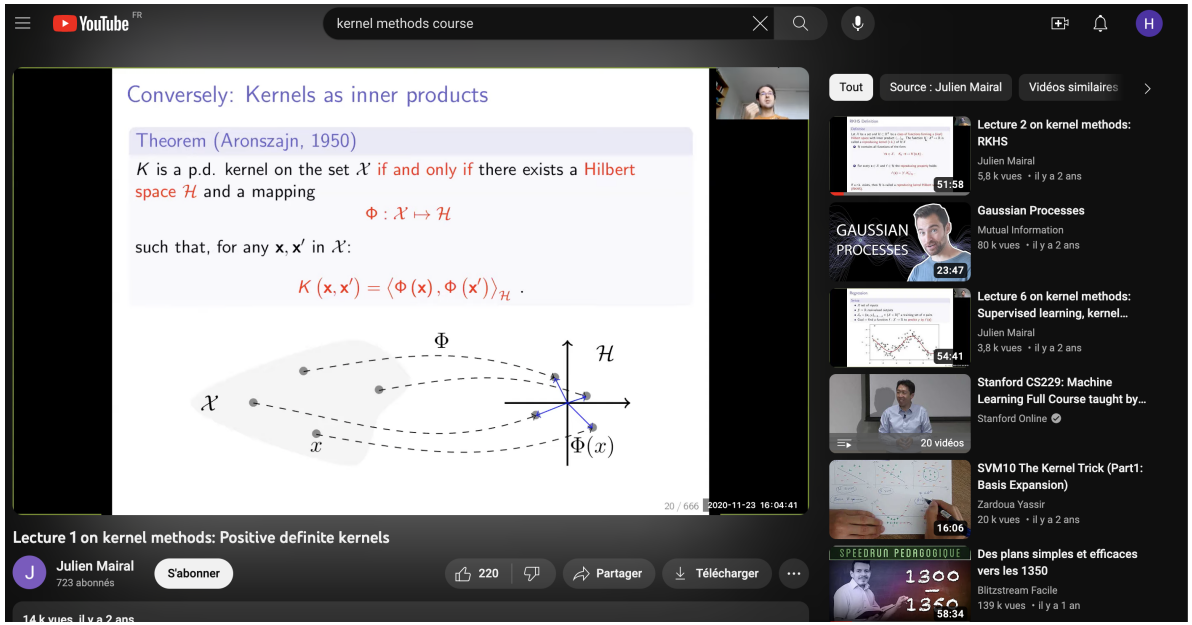
- Need for **continuous**-action methods

10 to 100 million auctions per second in all Criteo servers!

- Need for **efficient** and **scalable** learning methods

Example 2: YouTube Recommendation

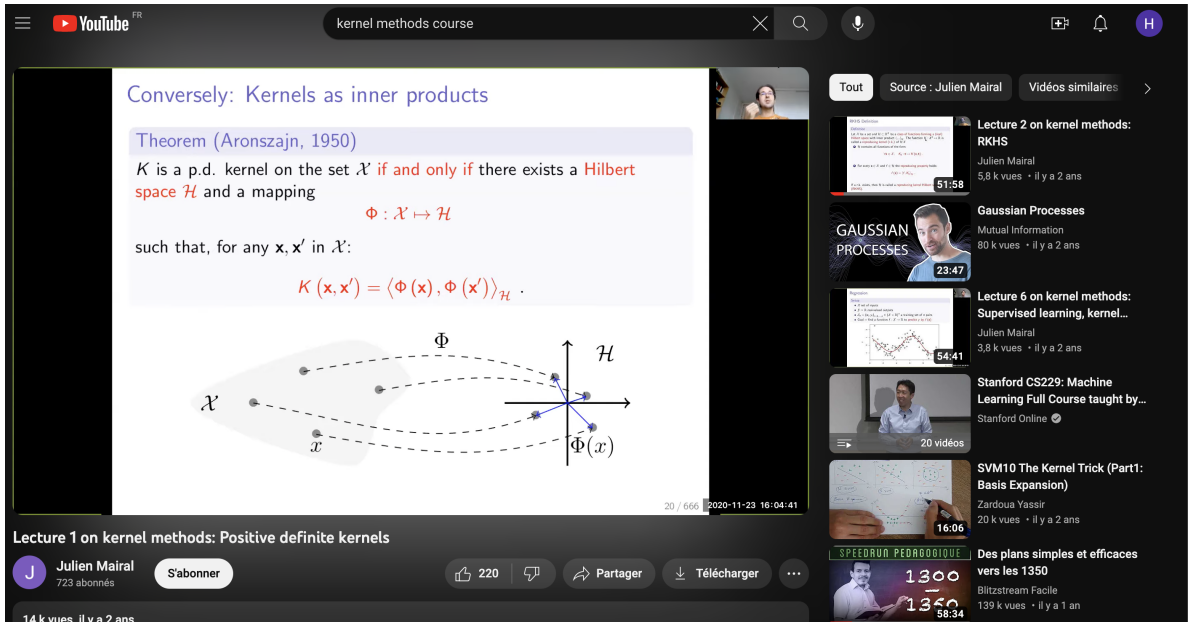
The recommendation task aims at proposing to a user the best item for them.



YouTube recommendation

Example 2: YouTube Recommendation

The recommendation task aims at proposing to a user the best item for them.

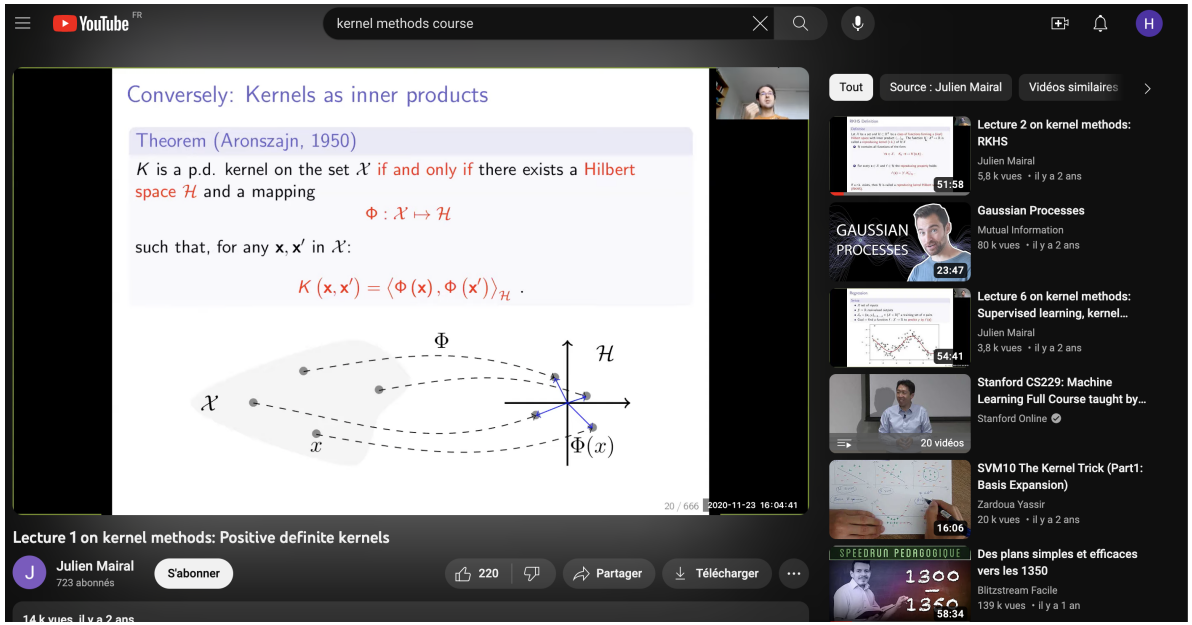


YouTube recommendation

- **context**: current video, user demographics, past interactions

Example 2: YouTube Recommendation

The recommendation task aims at proposing to a user the best item for them.

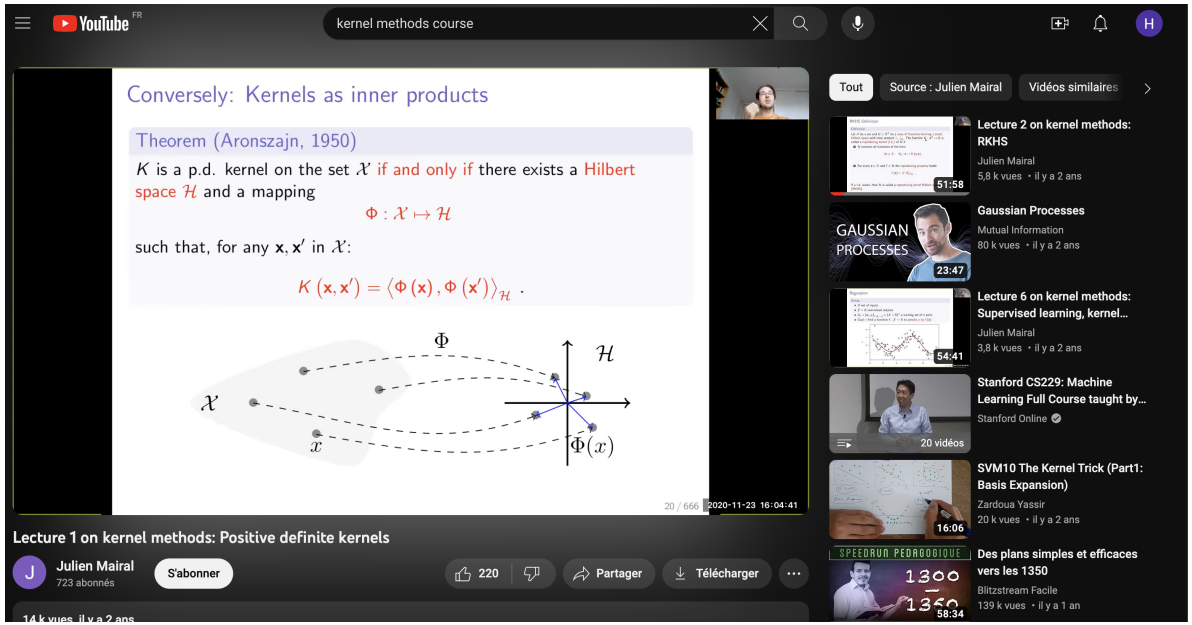


YouTube recommendation

- **context**: current video, user demographics, past interactions
- **action**: a ranking of recommended videos

Example 2: YouTube Recommendation

The recommendation task aims at proposing to a user the best item for them.

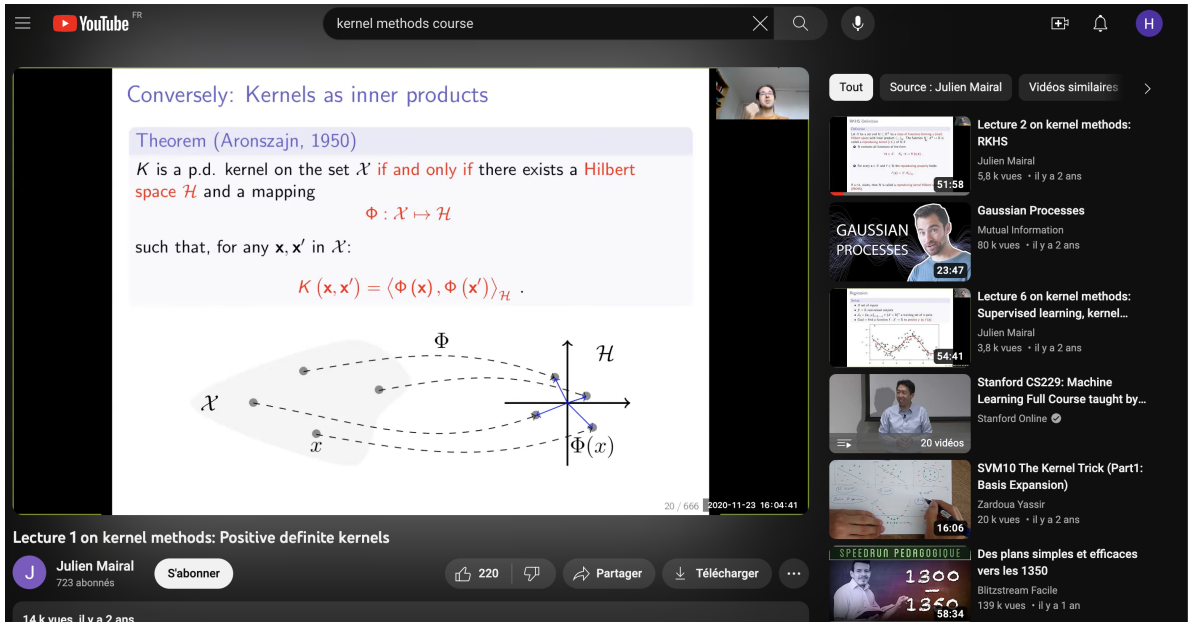


YouTube recommendation

- **context**: current video, user demographics, past interactions
- **action**: a ranking of recommended videos
- **feedback**: click, downstream time, etc..

Example 2: YouTube Recommendation

The recommendation task aims at proposing to a user the best item for them.



YouTube recommendation

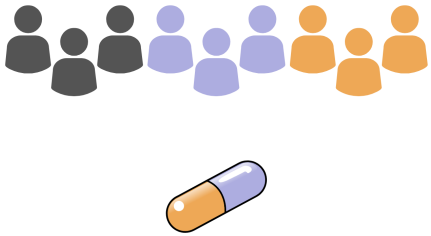
- **context**: current video, user demographics, past interactions
- **action**: a ranking of recommended videos
- **feedback**: click, downstream time, etc..

800 millions videos on YouTube, estimated 1 billion hours watched per day.

- **Large amounts** of log data

Example 3: Personalized Medicine

The task aims at assigning to groups of patients the best treatment.



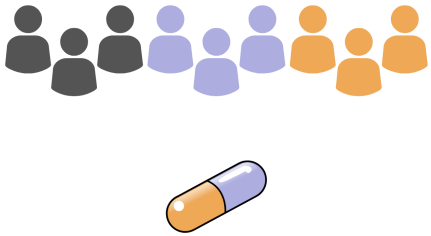
Single treatment *(one-fits-all)*



Personalized treatment

Example 3: Personalized Medicine

The task aims at assigning to groups of patients the best treatment.



Single treatment *(one-fits-all)*

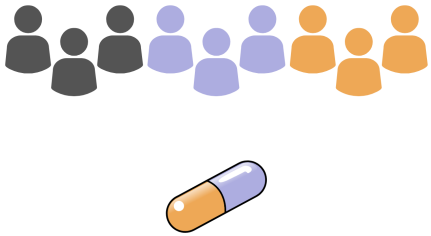


Personalized treatment

- **context**: patient info, genetic and biomarker analysis, life style past interactions

Example 3: Personalized Medicine

The task aims at assigning to groups of patients the best treatment.



Single treatment *(one-fits-all)*

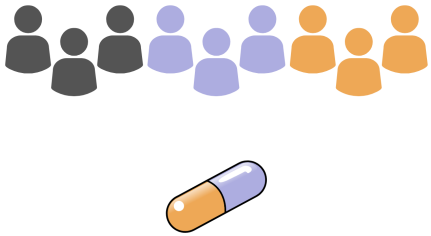


Personalized treatment

- **context**: patient info, genetic and biomarker analysis, life style past interactions
- **action**: treatment molecule, dose

Example 3: Personalized Medicine

The task aims at assigning to groups of patients the best treatment.



Single treatment *(one-fits-all)*

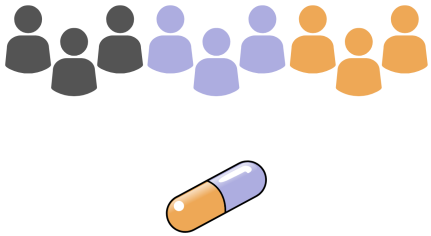


Personalized treatment

- **context**: patient info, genetic and biomarker analysis, life style past interactions
- **action**: treatment molecule, dose
- **feedback**: patient recovery

Example 3: Personalized Medicine

The task aims at assigning to groups of patients the best treatment.



Single treatment *(one-fits-all)*



Personalized treatment

- **context**: patient info, genetic and biomarker analysis, life style past interactions
- **action**: treatment molecule, dose
- **feedback**: patient recovery

Hard to hire patients for clinical trials, exploration can be unethical.

- Need for **adaptive** experiments and **safe** methods.

Disclaimer: personalized medicine is often not used in real-world medical applications because of legal issues related to exploration and safety.

Challenges in efficient sequential decision making

Online learning:

Design and analyze scalable learning methods for algorithms that work very well in practice.

Online learning:

Design and analyze scalable learning methods for algorithms that work very well in practice.

When?

Challenges in efficient sequential decision making

Online learning:

Design and analyze scalable learning methods for algorithms that work very well in practice.

When? Applications with mild risk in exploration.

Offline learning:

Variance reduction, effective modeling (applications with continuous action), large scale benchmarks and offline evaluation protocols.

Challenges in efficient sequential decision making

Online learning:

Design and analyze scalable learning methods for algorithms that work very well in practice.

When? Applications with mild risk in exploration.

Offline learning:

Variance reduction, effective modeling (applications with continuous action), large scale benchmarks and offline evaluation protocols.

When?

Challenges in efficient sequential decision making

Online learning:

Design and analyze scalable learning methods for algorithms that work very well in practice.

When? Applications with mild risk in exploration.

Offline learning:

Variance reduction, effective modeling (applications with continuous action), large scale benchmarks and offline evaluation protocols.

When? Sensitive applications.

Challenges in efficient sequential decision making

Online learning:

Design and analyze scalable learning methods for algorithms that work very well in practice.

When? Applications with mild risk in exploration.

Offline learning:

Variance reduction, effective modeling (applications with continuous action), large scale benchmarks and offline evaluation protocols.

When? Sensitive applications.

Offline/Online learning: batch feedback

Design and analyze an efficient batch learning strategy that bridge the gap between the two worlds.

Challenges in efficient sequential decision making

Online learning:

Design and analyze scalable learning methods for algorithms that work very well in practice.

When? Applications with mild risk in exploration.

Offline learning:

Variance reduction, effective modeling (applications with continuous action), large scale benchmarks and offline evaluation protocols.

When? Sensitive applications.

Offline/Online learning: batch feedback

Design and analyze an efficient batch learning strategy that bridge the gap between the two worlds.

When?

Challenges in efficient sequential decision making

Online learning:

Design and analyze scalable learning methods for algorithms that work very well in practice.

When? Applications with mild risk in exploration.

Offline learning:

Variance reduction, effective modeling (applications with continuous action), large scale benchmarks and offline evaluation protocols.

When? Sensitive applications.

Offline/Online learning: batch feedback

Design and analyze an efficient batch learning strategy that bridge the gap between the two worlds.

When? Whenever multiple deployments are allowed.

Challenges in efficient sequential decision making

Online learning:

Design and analyze scalable learning methods for algorithms that work very well in practice.

When? Applications with mild risk in exploration.

Offline learning:

Variance reduction, effective modeling (applications with continuous action), large scale benchmarks and offline evaluation protocols.

When? Sensitive applications.

Offline/Online learning: batch feedback

Design and analyze an efficient batch learning strategy that bridge the gap between the two worlds.

When? Whenever multiple deployments are allowed.

Focus on **effectiveness** and **efficiency** of the methods.

Part I - Efficient methods in sequential learning

Contributions

Zenati, H., Bietti, A., Diemert, E., Mairal, J., Martin, M., & Gaillard, P. (2022).Efficient kernelized UCB for contextual bandits. *International Conference on Artificial Intelligence and Statistics (AISTATS)*

Martin, M., Mertikopoulos, P., Rahier, T., & Zenati, H. (2022).Nested bandits. *International Conference on Machine Learning (ICML)*

Part I - Efficient methods in sequential learning

Contributions

Zenati, H., Bietti, A., Diemert, E., Mairal, J., Martin, M., & Gaillard, P. (2022).Efficient kernelized UCB for contextual bandits. *International Conference on Artificial Intelligence and Statistics (AISTATS)*

Martin, M., Mertikopoulos, P., Rahier, T., & Zenati, H. (2022).Nested bandits. *International Conference on Machine Learning (ICML)*

Contextual Bandits

At each round t, a bandit **agent** interacts with the environment:

- observes a **context** $x_t \in \mathcal{X}$, and $x_t \sim \mathcal{P}_{\mathcal{X}}$,
- takes an **action** $a_t \in \mathcal{A}$,
- receives a **reward** (outcome) $r_t = r(x_t, a_t) \sim \mathcal{P}_{\mathcal{R}}(\cdot|x_t, a_t)$

Important: we assume the reward to be drawn from a stochastic distribution.

At each round t, a bandit **agent** interacts with the environment:

- observes a **context** $x_t \in \mathcal{X}$, and $x_t \sim \mathcal{P}_{\mathcal{X}}$,
- takes an **action** $a_t \in \mathcal{A}$,
- receives a **reward** (outcome) $r_t = r(x_t, a_t) \sim \mathcal{P}_{\mathcal{R}}(\cdot|x_t, a_t)$

Important: we assume the reward to be drawn from a stochastic distribution.

The goal is to find an **agent** (policy) that **minimizes** the **regret**, over any horizon T:

$$R_T := \mathbb{E}\left[\sum_{t=1}^{T} \max_{a \in \mathcal{A}} r(x_t, a) - \sum_{t=1}^{T} r_t\right].$$

At each round t, a bandit **agent** interacts with the environment:

- observes a **context** $x_t \in \mathcal{X}$, and $x_t \sim \mathcal{P}_{\mathcal{X}}$,
- takes an **action** $a_t \in \mathcal{A}$,
- receives a **reward** (outcome) $r_t = r(x_t, a_t) \sim \mathcal{P}_{\mathcal{R}}(\cdot|x_t, a_t)$

Important: we assume the reward to be drawn from a stochastic distribution.

The goal is to find an **agent** (policy) that **minimizes** the **regret**, over any horizon T:

$$R_T := \mathbb{E}\left[\sum_{t=1}^{T} \max_{a \in \mathcal{A}} r(x_t, a) - \sum_{t=1}^{T} r_t\right].$$

Well known methods include ϵ-greedy, RegCB, SquareCB (Dylan J. Foster, 2020; Naoki Abe, 1999), and more simply Thompson sampling and OFUL-learning methods (Abbasi-yadkori et al., 2011).

Kernel UCB

In the Kernel UCB setting, we assume

$$r_t = \langle \theta^*, \phi(x_t, a_t) \rangle + \epsilon_t \,,$$

where $\theta^* \in \mathcal{H}$ the RKHS, ϕ is a feature map associated to $\mathcal{H}$ and the kernel k.

$\Longrightarrow$ **expressive** embeddings through the **joint embedding of context and actions** with the feature map.

Kernel UCB

In the Kernel UCB setting, we assume

$$r_t = \langle \theta^*, \phi(x_t, a_t) \rangle + \epsilon_t \,,$$

where $\theta^* \in \mathcal{H}$ the RKHS, ϕ is a feature map associated to $\mathcal{H}$ and the kernel k.

$\implies$ **expressive** embeddings through the **joint embedding of context and actions** with the feature map.

The algorithm estimates θ^* with a regularized least squares:

$$\hat{\theta}_t \in \operatorname*{arg\,min}_{\theta \in \mathcal{H}} \left\{ \sum_{s=1}^{t} \left(\langle \theta, \phi(x_s, a_s) \rangle - r_s \right)^2 + \lambda \|\theta\|^2 \right\}.$$

Kernel UCB

In the Kernel UCB setting, we assume

$$r_t = \langle \theta^*, \phi(x_t, a_t) \rangle + \epsilon_t \,,$$

where $\theta^* \in \mathcal{H}$ the RKHS, ϕ is a feature map associated to $\mathcal{H}$ and the kernel k.

$\implies$ **expressive** embeddings through the **joint embedding of context and actions** with the feature map.

The algorithm estimates θ^* with a regularized least squares:

$$\hat{\theta}_t \in \operatorname*{arg\,min}_{\theta \in \mathcal{H}} \left\{ \sum_{s=1}^{t} \left(\langle \theta, \phi(x_s, a_s) \rangle - r_s \right)^2 + \lambda \|\theta\|^2 \right\}.$$

Then, K-UCB builds a **confidence set** around the estimate $\hat{\theta}_t$.

$$\mathcal{C}_t = \{ \theta \in \mathcal{H} : \|\theta - \hat{\theta}_{t-1}\|_{V_{t-1}} \leq \beta \},$$

where V_t is a data-dependent regularized covariate matrix.

Kernel UCB

In the Kernel UCB setting, we assume

$$r_t = \langle \theta^*, \phi(x_t, a_t) \rangle + \epsilon_t \,,$$

where $\theta^* \in \mathcal{H}$ the RKHS, ϕ is a feature map associated to $\mathcal{H}$ and the kernel k.

$\implies$ **expressive** embeddings through the **joint embedding of context and actions** with the feature map.

The algorithm estimates θ^* with a regularized least squares:

$$\hat{\theta}_t \in \arg\min_{\theta \in \mathcal{H}} \left\{ \sum_{s=1}^{t} \left(\langle \theta, \phi(x_s, a_s) \rangle - r_s \right)^2 + \lambda \|\theta\|^2 \right\}.$$

Then, K-UCB builds a **confidence set** around the estimate $\hat{\theta}_t$.

$$\mathcal{C}_t = \{ \theta \in \mathcal{H} : \|\theta - \hat{\theta}_{t-1}\|_{V_{t-1}} \leq \beta \},$$

where V_t is a data-dependent regularized covariate matrix.

The algorithm takes the upper bound on $\mathcal{C}_t$ and then selects the **best action**:

$$\text{K-UCB}_t(a) = \max_{\theta \in \mathcal{C}_t} \langle \theta, \phi(x_t, a) \rangle, \quad a_t \in \arg\max_{a \in \mathcal{A}} \text{K-UCB}_t(a)$$

Proposition (K-UCB regret (Zenati et al., 2022))

The algorithm enjoys the following pseudo-regret bound:

$$R_T \lesssim \sqrt{T}\left(\|\theta^*\|\sqrt{\lambda d_{\text{eff}}} + d_{\text{eff}}\right),$$

d_{eff} is the effective dimension associated to λ and the kernel matrix $K_t = [\langle \phi(x_i, a_i), \phi(x_j, a_j)\rangle]_{1\leq i,j\leq t}$.

$$d_{\text{eff}}(\lambda, T) = \text{Tr}(K_T(K_T + \lambda I_T)^{-1}).$$

Proposition (K-UCB regret (Zenati et al., 2022))

The algorithm enjoys the following pseudo-regret bound:

$$R_T \lesssim \sqrt{T}\left(\|\theta^*\|\sqrt{\lambda d_{\text{eff}}} + d_{\text{eff}}\right),$$

d_{eff} is the effective dimension associated to λ and the kernel matrix $K_t = [\langle \phi(x_i, a_i), \phi(x_j, a_j)\rangle]_{1\leq i,j\leq t}$.

$$d_{\text{eff}}(\lambda, T) = \text{Tr}(K_T(K_T + \lambda I_T)^{-1}).$$

Choose $a_t \leftarrow \arg\max_{a\in\mathcal{A}} \hat{\mu}_{t,a} + \beta_t \hat{\sigma}_{t,a}$

$\hat{\mu}_{t,a} \leftarrow K_{\mathcal{S}_{t-1}}(s_{t,a})^\top (K_t + \lambda I_t)^{-1} Y_{t-1}$

$\hat{\sigma}^2_{t,a} \leftarrow \frac{1}{\lambda} k(s_{t,a}, s_{t,a}) - \frac{1}{\lambda} K_{\mathcal{S}_{t-1}}(s_{t,a})^\top (K_t + \lambda I_t)^{-1} K_{\mathcal{S}_{t-1}}(s_{t,a})$

where $s_{t,a} = (x_t, a)$, and let $K_t(s) = [k((x_1, a_1), s), \dots k((x_t, a_t), s)]^\top$

Proposition (K-UCB regret (Zenati et al., 2022))

The algorithm enjoys the following pseudo-regret bound:

$$R_T \lesssim \sqrt{T}\left(\|\theta^*\|\sqrt{\lambda d_{\text{eff}}} + d_{\text{eff}}\right),$$

d_{eff} is the effective dimension associated to λ and the kernel matrix $K_t = [\langle \phi(x_i, a_i), \phi(x_j, a_j)\rangle]_{1\leq i,j\leq t}$.

$$d_{\text{eff}}(\lambda, T) = \text{Tr}(K_T(K_T + \lambda I_T)^{-1}).$$

Choose $a_t \leftarrow \arg\max_{a\in\mathcal{A}} \hat{\mu}_{t,a} + \beta_t \hat{\sigma}_{t,a}$

$\hat{\mu}_{t,a} \leftarrow K_{\mathcal{S}_{t-1}}(s_{t,a})^\top \ (K_t + \lambda I_t)^{-1} \ Y_{t-1}$

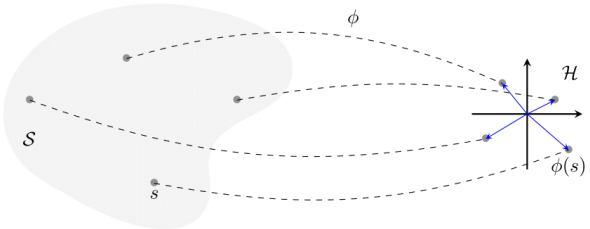
$\hat{\sigma}^2_{t,a} \leftarrow \frac{1}{\lambda} k(s_{t,a}, s_{t,a}) - \frac{1}{\lambda} K_{\mathcal{S}_{t-1}}(s_{t,a})^\top \ (K_t + \lambda I_t)^{-1} \ K_{\mathcal{S}_{t-1}}(s_{t,a})$

where $s_{t,a} = (x_t, a)$, and let $K_t(s) = [k((x_1, a_1), s), \dots k((x_t, a_t), s)]^\top$

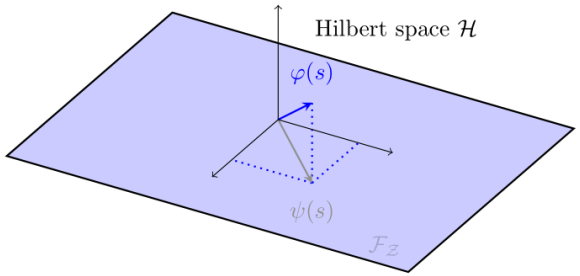
Computational complexity

The algorithm runs in $\mathcal{O}(T^3)$ **time complexity**.

Incremental approximations of the RKHS (Calandriello et al., 2017)



Embedding in the RKHS $\mathcal{H}$

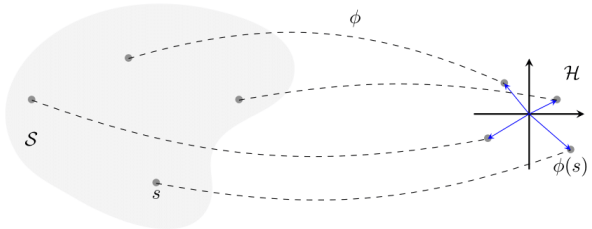


Nyström projection of a feature map φ

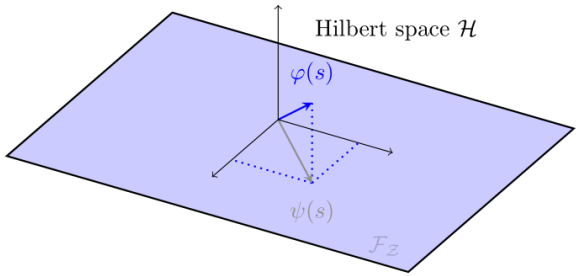
Build a dictionary $\mathcal{Z}$ of historical observations to build a subspace of the RKHS:

$$\mathcal{F}_{\mathcal{Z}} = \mathrm{Span}_{z \in \mathcal{Z}}(\phi(z))$$

Incremental approximations of the RKHS (Calandriello et al., 2017)



Embedding in the RKHS $\mathcal{H}$



Nyström projection of a feature map φ

Build a dictionary $\mathcal{Z}$ of historical observations to build a subspace of the RKHS:

$$\mathcal{F}_{\mathcal{Z}} = \text{Span}_{z \in \mathcal{Z}}(\phi(z))$$

Algorithm Incremental KORS subroutine

Input: Past dictionary $\mathcal{Z}$, context-action s_t, regularization μ, budget γ
Compute the leverage score $\tilde{\tau}_t$ from $\mathcal{Z}, s_t, \mu, \epsilon$
  Compute $\tilde{p}_t = \min\{\gamma \tilde{\tau}_t, 1\}$
  Draw $z_t \sim \mathcal{B}(\tilde{p}_t)$ and if $z_t = 1$, add s_t to $\mathcal{Z}$

Proposition

The sequence of dictionaries $\mathcal{Z}_1 \subset \mathcal{Z}_2 \subset \ldots \mathcal{Z}_T$ learned by KORS with parameters $\mu > 0$ satisfies with high probability that $m = |\mathcal{Z}_t| \approx d_{\text{eff}}(\mu, T)$.

Efficient Kernel UCB regret bound (Zenati et al., 2022)

EK-UCB estimates the parameter θ^* with a **projected** parameter $\tilde{\theta}_t$ to build a confidence set $\tilde{\mathcal{C}}_t$:

$$\tilde{\theta}_t \in \underset{\theta \in \mathcal{F}_{\mathcal{Z}}}{\arg\min} \left\{ \sum_{s=1}^{t} \left(\langle \theta, \phi(x_s, a_s) \rangle - r_s \right)^2 + \lambda \|\theta\|^2 \right\}.$$

Efficient Kernel UCB regret bound (Zenati et al., 2022)

EK-UCB estimates the parameter θ^* with a **projected** parameter $\tilde{\theta}_t$ to build a confidence set $\tilde{\mathcal{C}}_t$:

$$\tilde{\theta}_t \in \operatorname*{arg\,min}_{\theta \in \mathcal{F}_{\mathcal{Z}}} \left\{ \sum_{s=1}^{t} \left(\langle \theta, \phi(x_s, a_s) \rangle - r_s \right)^2 + \lambda \|\theta\|^2 \right\}.$$

Theorem (EK-UCB regret)

Writing $m := |\mathcal{Z}_T|$, the regret of the EK-UCB algorithm matches

$$R_T \lesssim \sqrt{T} \left(\sqrt{\frac{\mu m}{\lambda}} + \sqrt{d_{\text{eff}}} \right) \left(\|\theta^*\| (\sqrt{\lambda} + \sqrt{\mu}) + \sqrt{d_{\text{eff}}} \right).$$

When choosing $\mu = \lambda$ we have $m \lesssim d_{\text{eff}}$ and

$$R_T \lesssim \sqrt{T} \left(\|\theta^*\| \sqrt{\lambda d_{\text{eff}}} + d_{\text{eff}} \right).$$

Efficient Kernel UCB regret bound (Zenati et al., 2022)

EK-UCB estimates the parameter θ^* with a **projected** parameter $\tilde{\theta}_t$ to build a confidence set $\tilde{\mathcal{C}}_t$:

$$\tilde{\theta}_t \in \underset{\theta \in \mathcal{F}_{\mathcal{Z}}}{\arg\min} \left\{ \sum_{s=1}^{t} \left(\langle \theta, \phi(x_s, a_s) \rangle - r_s \right)^2 + \lambda \|\theta\|^2 \right\}.$$

Theorem (EK-UCB regret)

Writing $m := |\mathcal{Z}_T|$, the regret of the EK-UCB algorithm matches

$$R_T \lesssim \sqrt{T} \left(\sqrt{\frac{\mu m}{\lambda}} + \sqrt{d_{\text{eff}}} \right) \left(\|\theta^*\| (\sqrt{\lambda} + \sqrt{\mu}) + \sqrt{d_{\text{eff}}} \right).$$

When choosing $\mu = \lambda$ we have $m \lesssim d_{\text{eff}}$ and

$$R_T \lesssim \sqrt{T} \left(\|\theta^*\| \sqrt{\lambda d_{\text{eff}}} + d_{\text{eff}} \right).$$

In the update rule, we now only invert a matrix of size $m \times m$, with $m \approx d_{\text{eff}}$.

Computational complexity

The algorithm runs in $\mathcal{O}(T d_{\text{eff}}^2)$ **time complexity**.

Related Work

Previous works BKB (Calandriello et al., 2019) and BBKB (Calandriello et al., 2020) considered a **non contextual** setting.

Related Work

Previous works BKB (Calandriello et al., 2019) and BBKB (Calandriello et al., 2020) considered a **non contextual** setting.

To obtain efficient algorithms, they recompute the Nyström dictionary based on **all samples**.

Related Work

Previous works BKB (Calandriello et al., 2019) and BBKB (Calandriello et al., 2020) considered a **non contextual** setting.

To obtain efficient algorithms, they recompute the Nyström dictionary based on **all samples**.

Instead, we considered a contextual setting and made an analysis to use an **incremental** projection scheme.

Related Work

Previous works BKB (Calandriello et al., 2019) and BBKB (Calandriello et al., 2020) considered a **non contextual** setting.

To obtain efficient algorithms, they recompute the Nyström dictionary based on **all samples**.

Instead, we considered a contextual setting and made an analysis to use an **incremental** projection scheme.

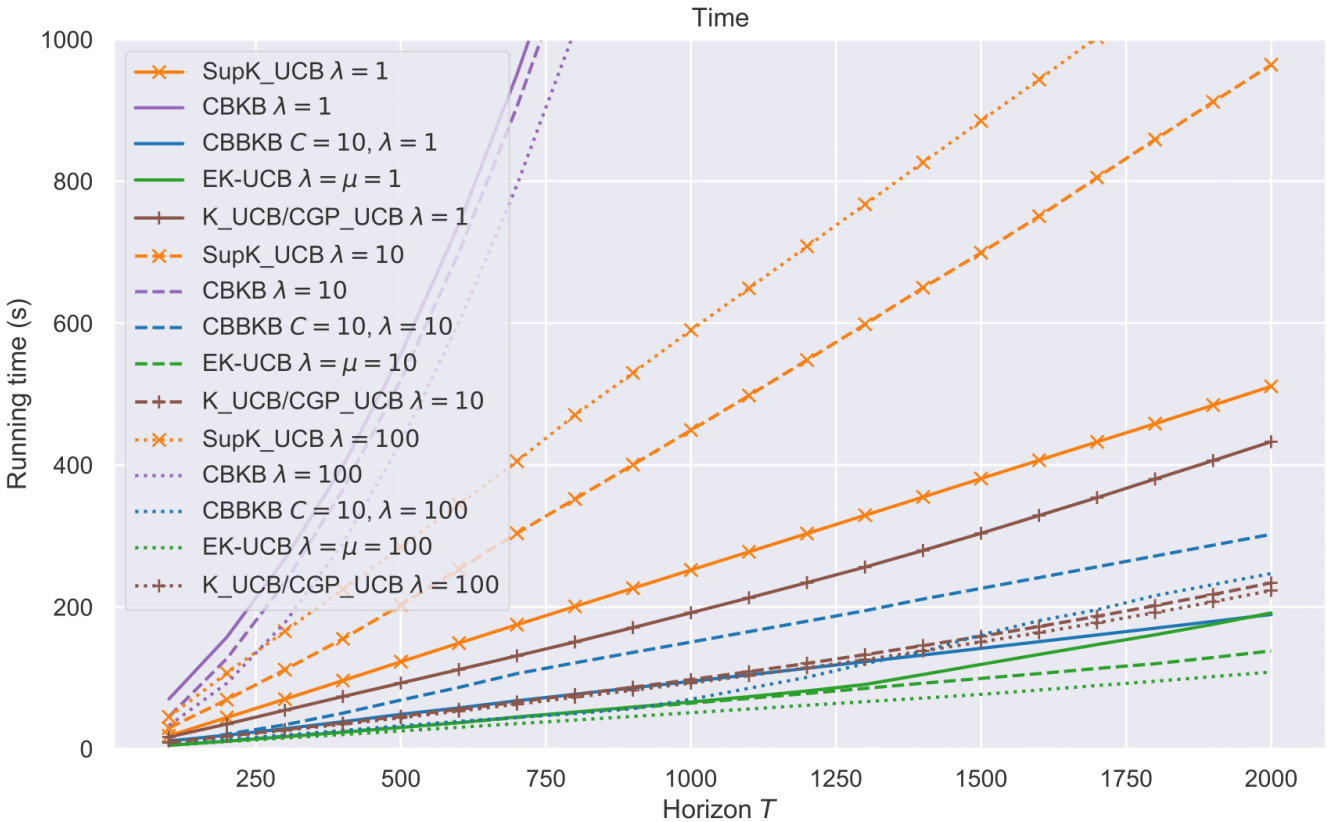
Algorithm	Time Complexity
CGP-UCB (Krause & Ong, 2011)	$\mathcal{O}(T^3)$
SupKernelUCB (Valko et al., 2013)	$\mathcal{O}(T^3)$
K-UCB (ours)	$\mathcal{O}(T^3)$
C-BKB (Calandriello et al., 2019)	$\mathcal{O}\left(T^2 d_{\text{eff}}^2\right)$
C-BBKB (Calandriello et al., 2020)	$\mathcal{O}\left(T d_{\text{eff}}^3\right)$
EK-UCB (ours)	$\mathcal{O}(T d_{\text{eff}}^2)$

Comparison of regrets, space and time complexities

$\implies$ The incremental projection updates allow to perform **more efficient** approximations in the joint context-action space.

Numerical Experiments (Zenati et al., 2022)

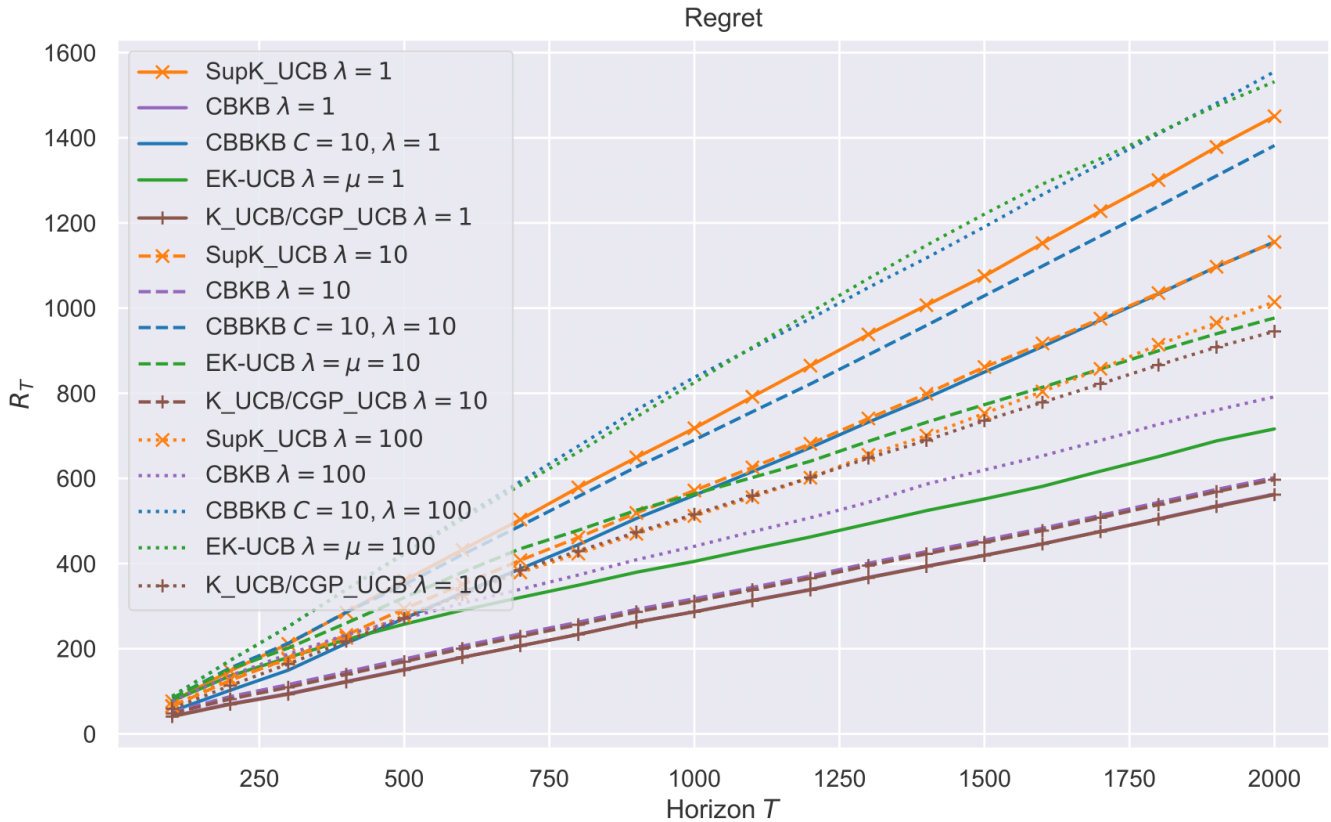
Experiment on synthetic environments with contexts to compare to K-UCB, SupK-UCB, CBKB and CBBKB.



- EK-UCB (green) has the best computational complexity: smaller projection μ induce a better regret but a higher time complexity.
- The practical computational complexity of "efficient" BKB can be worse than the original algorithm!

Numerical Experiments (Zenati et al., 2022)

Experiment on synthetic environments with contexts to compare to K-UCB, SupK-UCB, CBKB and CBBKB.



- EK-UCB (green) has the best computational complexity: smaller projection μ induce a better regret but a higher time complexity.
- The practical computational complexity of "efficient" BKB can be worse than the original algorithm!
- EK-UCB with parameters $\mu = 1, 10$ have low regret.

Discussions

- The EK-UCB algorithm runs in $\mathcal{O}(Td_{\text{eff}}^2)$ time complexity, which significantly improves over the standard contextual kernel UCB method.
- The **incremental** projection updates are crucial to perform **efficient approximations** in the joint context-action space and are more efficient than the dictionary resampling strategy.

Discussions

- The EK-UCB algorithm runs in $\mathcal{O}(Td_{\text{eff}}^2)$ time complexity, which significantly improves over the standard contextual kernel UCB method.
- The **incremental** projection updates are crucial to perform **efficient approximations** in the joint context-action space and are more efficient than the dictionary resampling strategy.

But the **stochastic** assumptions we make on the environment may be too strong for some applications!

Discussions

- The EK-UCB algorithm runs in $\mathcal{O}(Td_{\text{eff}}^2)$ time complexity, which significantly improves over the standard contextual kernel UCB method.
- The **incremental** projection updates are crucial to perform **efficient approximations** in the joint context-action space and are more efficient than the dictionary resampling strategy.

But the **stochastic** assumptions we make on the environment may be too strong for some applications!

In some applications (at Criteo especially), an **adversary** can learn the behavior of the agent and adapt their strategy.

Discussions

- The EK-UCB algorithm runs in $\mathcal{O}(Td_{\text{eff}}^2)$ time complexity, which significantly improves over the standard contextual kernel UCB method.
- The **incremental** projection updates are crucial to perform **efficient approximations** in the joint context-action space and are more efficient than the dictionary resampling strategy.

But the **stochastic** assumptions we make on the environment may be too strong for some applications!

In some applications (at Criteo especially), an **adversary** can learn the behavior of the agent and adapt their strategy.

What if we consider that setting?

Another work: nested bandits

In sequential decision making, consider the adversarial setting with k arms and a horizon T: Exponential Weights algorithms (Littlestone & Warmuth, 1994) suffer:

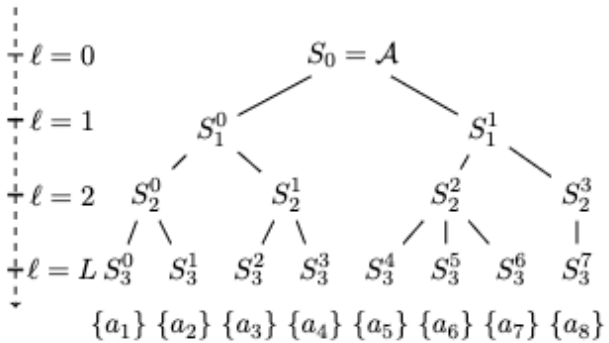
$$R_T \leq \sqrt{2k \log(k) T}.$$

Another work: nested bandits

In sequential decision making, consider the adversarial setting with k arms and a horizon T: Exponential Weights algorithms (Littlestone & Warmuth, 1994) suffer:

$$R_T \leq \sqrt{2k \log(k) T}.$$

Problem: if k is very large but some alternatives have a reward similarity structure: Can we do better?



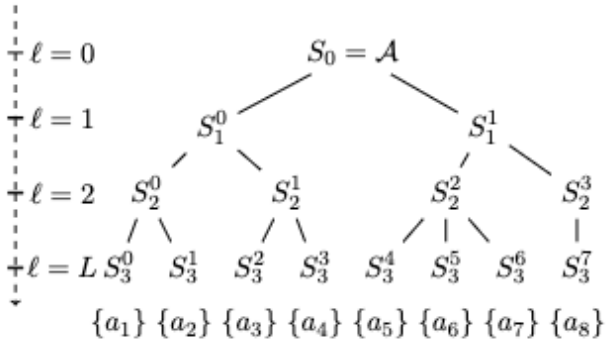
"Nested" similarity structure

Another work: nested bandits

In sequential decision making, consider the adversarial setting with k arms and a horizon T: Exponential Weights algorithms (Littlestone & Warmuth, 1994) suffer:

$$R_T \leq \sqrt{2k \log(k) T}.$$

Problem: if k is very large but some alternatives have a reward similarity structure: Can we do better?



"Nested" similarity structure

We propose **Nested Exponential Weights**:

$$R_T \leq \sqrt{2k_{\text{eff}} \log(k) T} \quad \text{and} \quad k_{\text{eff}} \ll k.$$

Martin, M., Mertikopoulos, P., Rahier, T., & Zenati, H. (2022).Nested bandits. *International Conference on Machine Learning (ICML)*

Part II - Efficient Counterfactual Learning Methods

Contributions

Zenati, H., Bietti, A., Martin, M., Diemert, E., & Mairal, J. (2020).Optimization approaches for counterfactual risk minimization with continuous actions. *International Conference on Learning Representation (ICLR), Causal Learning for Decision Making Workshop*

Zenati, H., Bietti, A., Martin, M., Diemert, E., Gaillard, P., & Mairal, J. (2020).Counterfactual learning of stochastic policies with continuous actions: From models to offline evaluation. https://arxiv.org/abs/2004.11722

Zenati, H., Diemert, E., Martin, M., Mairal, J., & Gaillard, P. (2023).Sequential counterfactual risk minimization. *International Conference on Machine Learning (ICML)*

Part II - Efficient Counterfactual Learning Methods

Contributions

Zenati, H., Bietti, A., Martin, M., Diemert, E., & Mairal, J. (2020).Optimization approaches for counterfactual risk minimization with continuous actions. *International Conference on Learning Representation (ICLR), Causal Learning for Decision Making Workshop*

Zenati, H., Bietti, A., Martin, M., Diemert, E., Gaillard, P., & Mairal, J. (2020).Counterfactual learning of stochastic policies with continuous actions: From models to offline evaluation. https://arxiv.org/abs/2004.11722

Zenati, H., Diemert, E., Martin, M., Mairal, J., & Gaillard, P. (2023).Sequential counterfactual risk minimization. *International Conference on Machine Learning (ICML)*

Setting

We are given an observational dataset $\mathcal{D}_0 = (x_i, a_i, y_i)_{i=1,\dots,n}$, with:

Setting

We are given an observational dataset $\mathcal{D}_0 = (x_i, a_i, y_i)_{i=1,\dots,n}$, with:

- observed a **context** $x_i \in \mathcal{X}$, and $x_i \sim \mathcal{P}_{\mathcal{X}}$,

Setting

We are given an observational dataset $\mathcal{D}_0 = (x_i, a_i, y_i)_{i=1,\dots,n}$, with:

- observed a **context** $x_i \in \mathcal{X}$, and $x_i \sim \mathcal{P}_{\mathcal{X}}$,
- sampled an **action** (treatment) $a_i \in \mathcal{A}$ from a logging policy π_0 with propensity $\pi_{i,0} = \pi_0(a_i|x_i)$,

Setting

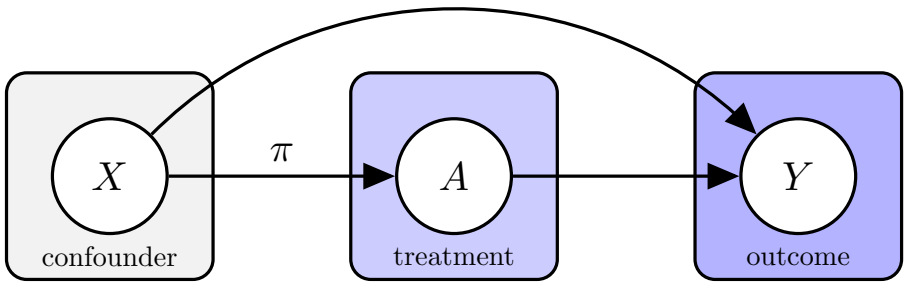
We are given an observational dataset $\mathcal{D}_0 = (x_i, a_i, y_i)_{i=1,\dots,n}$, with:

- observed a **context** $x_i \in \mathcal{X}$, and $x_i \sim \mathcal{P}_{\mathcal{X}}$,
- sampled an **action** (treatment) $a_i \in \mathcal{A}$ from a logging policy π_0 with propensity $\pi_{i,0} = \pi_0(a_i|x_i)$,
- observed a **loss** (outcome) $y_i \sim \mathcal{P}_{\mathcal{Y}}(\cdot|x_i, a_i)$

Setting

We are given an observational dataset $\mathcal{D}_0 = (x_i, a_i, y_i)_{i=1,\dots,n}$, with:

- observed a **context** $x_i \in \mathcal{X}$, and $x_i \sim \mathcal{P}_{\mathcal{X}}$,
- sampled an **action** (treatment) $a_i \in \mathcal{A}$ from a logging policy π_0 with propensity $\pi_{i,0} = \pi_0(a_i|x_i)$,
- observed a **loss** (outcome) $y_i \sim \mathcal{P}_{\mathcal{Y}}(\cdot|x_i, a_i)$



The goal is to find a policy that **minimizes** the **risk**, the expected loss of running π:

$$L(\pi) = \mathbb{E}_{x\sim\mathcal{P}_{\mathcal{X}}}\, \mathbb{E}_{a\sim\pi(\cdot|x)}\, \mathbb{E}_{y\sim\mathcal{P}_{\mathcal{Y}}(\cdot|x,a)}\, [y]$$

This requires off-policy evaluation (modelling + estimation) and optimization.

We consider **parametric** policies π_θ associated to a **model** $\theta \in \Theta$. The risk L of a **model** $\theta \in \Theta$ is given by:

$$L(\theta) = \mathbb{E}_{x \sim \mathcal{P}_\mathcal{X}} \, \mathbb{E}_{a \sim \pi_\theta(\cdot|x)} \, \mathbb{E}_{y \sim \mathcal{P}_\mathcal{Y}(\cdot|x,a)} \, [y]$$

Counterfactual Policy Evaluation

We consider **parametric** policies π_θ associated to a **model** $\theta \in \Theta$. The risk L of a **model** $\theta \in \Theta$ is given by:

$$L(\theta) = \mathbb{E}_{x \sim \mathcal{P}_\mathcal{X}} \, \mathbb{E}_{a \sim \pi_\theta(\cdot|x)} \, \mathbb{E}_{y \sim \mathcal{P}_\mathcal{Y}(\cdot|x,a)} \, [y]$$

Counterfactual question: *"What would have been the risk of the system, should the model θ_0 had been θ?"*

Counterfactual Policy Evaluation

We consider **parametric** policies π_θ associated to a **model** $\theta \in \Theta$. The risk L of a **model** $\theta \in \Theta$ is given by:

$$L(\theta) = \mathbb{E}_{x \sim \mathcal{P}_\mathcal{X}} \, \mathbb{E}_{a \sim \pi_\theta(\cdot|x)} \, \mathbb{E}_{y \sim \mathcal{P}_\mathcal{Y}(\cdot|x,a)} \, [y]$$

Counterfactual question: *"What would have been the risk of the system, should the model θ_0 had been θ?"*

We don't have access to samples drawn from the policy associated to model θ, rather we **observed samples** drawn from the **logging policy** associated to θ_0.

We consider **parametric** policies π_θ associated to a **model** $\theta \in \Theta$. The risk L of a **model** $\theta \in \Theta$ is given by:

$$L(\theta) = \mathbb{E}_{x\sim\mathcal{P}_\mathcal{X}}\, \mathbb{E}_{a\sim\pi_\theta(\cdot|x)}\, \mathbb{E}_{y\sim\mathcal{P}_\mathcal{Y}(\cdot|x,a)}\, [y]$$

Counterfactual question: *"What would have been the risk of the system, should the model θ_0 had been θ?"*

We don't have access to samples drawn from the policy associated to model θ, rather we **observed samples** drawn from the **logging policy** associated to θ_0.

By using the IPS method (Horvitz & Thompson, 1952):

$$L(\theta) = \mathbb{E}_{a\sim\pi_\theta(\cdot|x),x,y}\, [y] = \mathbb{E}_{a\sim\pi_{\theta_0}(\cdot|x),x,y}\, \left[y \frac{\pi_\theta(a|x)}{\pi_{\theta_0}(a|x)} \right]$$

Then, a natural estimator is

$$\hat{L}^{\text{IPS}}(\theta) = \frac{1}{n} \sum_{i=1}^{n} y_i \frac{\pi_\theta(a_i|x_i)}{\pi_{\theta_0}(a_i|x_i)}. \qquad \text{(IPS)}$$

Importance weighting has large variance, a clipped estimator has been proposed:

$$\hat{L}^{\text{cIPS}}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \underbrace{y_i \min\left\{\frac{\pi_\theta(a_i|x_i)}{\pi_{\theta_0}(a_i|x_i)}, \alpha\right\}}_{\nu_i(\theta)}, \qquad \text{(cIPS)}$$

where α is a clipping parameter.

Importance weighting has large variance, a clipped estimator has been proposed:

$$\hat{L}^{\text{cIPS}}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \underbrace{y_i \min\left\{\frac{\pi_\theta(a_i|x_i)}{\pi_{\theta_0}(a_i|x_i)}, \alpha\right\}}_{\nu_i(\theta)}, \qquad \text{(cIPS)}$$

where α is a clipping parameter.

Counterfactual Risk Minimization (CRM) Swaminathan and Joachims (2015) find a model θ that minimizes a regularized estimate of $L(\theta)$.

$$\hat{\theta}^{\text{CRM}} \in \arg\min_{\theta\in\Theta} \hat{L}^{\text{cIPS}}(\theta) + \lambda\sqrt{\frac{\widehat{\text{Var}}(\nu_i(\theta))}{n}} \quad \text{with} \quad \lambda > 0. \qquad \text{(CRM)}$$

The learning objective is conservative.

Other works in variance regularization

Regularizing the variance is an important problem in CRM:

Other works in variance regularization

Regularizing the variance is an important problem in CRM:

- Self-normalized estimator (SNIPS) (Owen, 2013):

$$\hat{L}^{\text{SNIPS}}(\theta) = \frac{\sum_{i=1}^{n} y_i w_i^{\theta}}{\sum_{i=1}^{n} w_i^{\theta}}, \quad w_i^{\theta} = \frac{\pi_\theta(a_i|x_i)}{\pi_{\theta_0}(a_i|x_i)} \qquad \text{(SNIPS)}$$

Other works in variance regularization

Regularizing the variance is an important problem in CRM:

- Self-normalized estimator (SNIPS) (Owen, 2013):

$$\hat{L}^{\text{SNIPS}}(\theta) = \frac{\sum_{i=1}^{n} y_i w_i^{\theta}}{\sum_{i=1}^{n} w_i^{\theta}}, \quad w_i^{\theta} = \frac{\pi_\theta(a_i|x_i)}{\pi_{\theta_0}(a_i|x_i)} \tag{SNIPS}$$

- Doubly robust estimator (DR) (Dudik et al., 2014):

$$\hat{L}^{\text{DR}}(\theta) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{\eta}(x_i, a_i))\,\frac{\pi_\theta(a_i|x_i)}{\pi_{\theta_0}(a_i|x_i)} + \frac{1}{n}\sum_{i=1}^{n}\sum_{a\in\mathcal{A}}\hat{\eta}(x_i, a)\pi_\theta(a|x_i), \tag{DR}$$

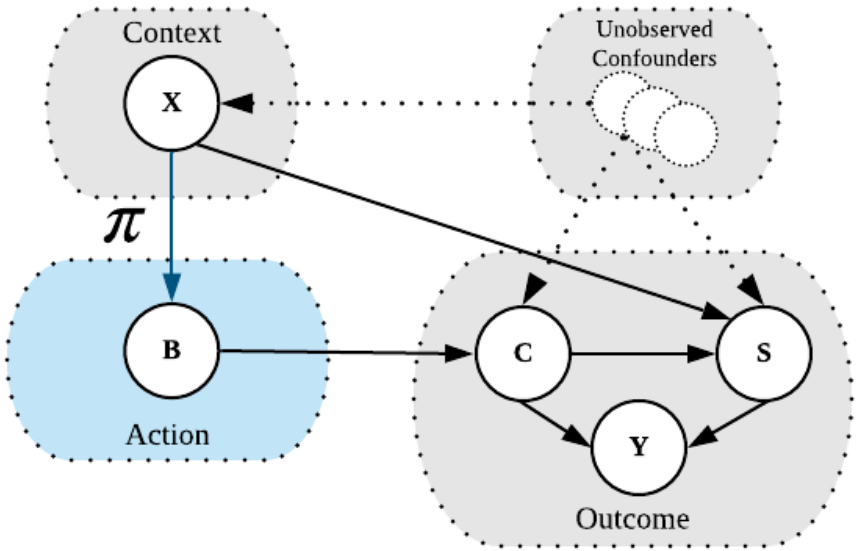
with a learned estimator $\hat{\eta}(x, a)$ of expected costs, for instance by using ridge regression to fit $y_i \approx \hat{\eta}(x_i, a_i)$ on the training data.

Continuous action problems, a new dataset: CoCoA (Zenati et al., 2020)

Lack of methods in the literature and benchmarks for continuous actions problems.

Continuous action problems, a new dataset: CoCoA (Zenati et al., 2020)

Lack of methods in the literature and benchmarks for continuous actions problems.



CoCoA causal graph

B is **bid**. The **cost** $c \sim C$ and a number of **sales** S defines a **loss** Y.

n	d	$\mathbb{E}[-Y]$	$\mathbb{V}[Y]$	$\mathbb{V}[A]$	$\mathbb{P}(Y \neq 0)$
120.10^6	3	11.34	9455	.01	.07

CoCoA dataset summary statistics.

Offline evaluation protocol (Zenati et al., 2020)

How to assess that a policy **improves** upon the logging policy?

Offline evaluation protocol (Zenati et al., 2020)

How to assess that a policy **improves** upon the logging policy?

- **Diagnose** when the offline estimation is too uncertain: use the effective sample size (Owen, 2013).

Offline evaluation protocol (Zenati et al., 2020)

How to assess that a policy **improves** upon the logging policy?

- **Diagnose** when the offline estimation is too uncertain: use the effective sample size (Owen, 2013).
- Build **confidence interval** around the offline estimation: use SNIPS estimates.

Offline evaluation protocol (Zenati et al., 2020)

How to assess that a policy **improves** upon the logging policy?

- **Diagnose** when the offline estimation is too uncertain: use the effective sample size (Owen, 2013).
- Build **confidence interval** around the offline estimation: use SNIPS estimates.

Still, it remains **difficult** to obtain results in real-world problems because the methods often fail to pass the offline evaluation protocol:

	Logging policy	(Chen et al., 2016)	(Kallus & Zhou, 2018)	Ours
CoCoA	11.34	11.03 ± 0.15	11.38 ± 0.07	$\mathbf{11.44}^{*} \pm 0.10$

Test rewards of continuous action methods: the higher the better

Offline evaluation protocol (Zenati et al., 2020)

How to assess that a policy **improves** upon the logging policy?

- **Diagnose** when the offline estimation is too uncertain: use the effective sample size (Owen, 2013).
- Build **confidence interval** around the offline estimation: use SNIPS estimates.

Still, it remains **difficult** to obtain results in real-world problems because the methods often fail to pass the offline evaluation protocol:

	Logging policy	(Chen et al., 2016)	(Kallus & Zhou, 2018)	Ours
CoCoA	11.34	11.03 ± 0.15	11.38 ± 0.07	$\mathbf{11.44}^{*} \pm 0.10$

Test rewards of continuous action methods: the higher the better

If the logging policy did not explore well the action space, offline learning becomes very difficult!

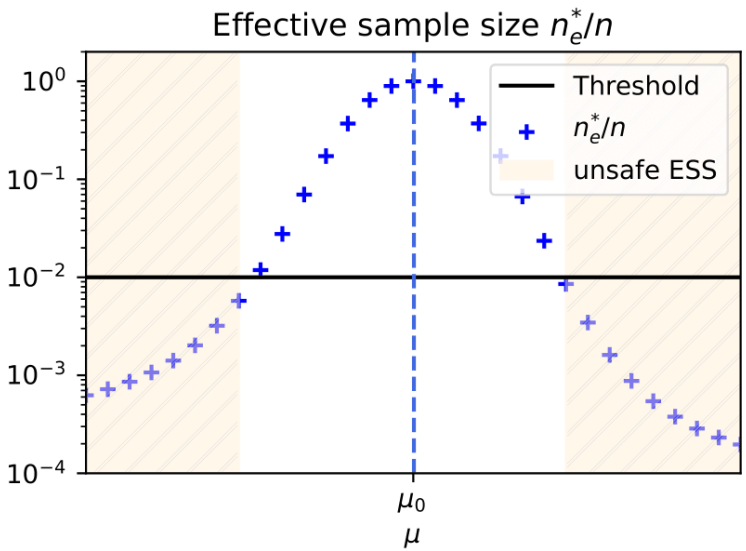
The underexploration problem (Zenati et al., 2023)

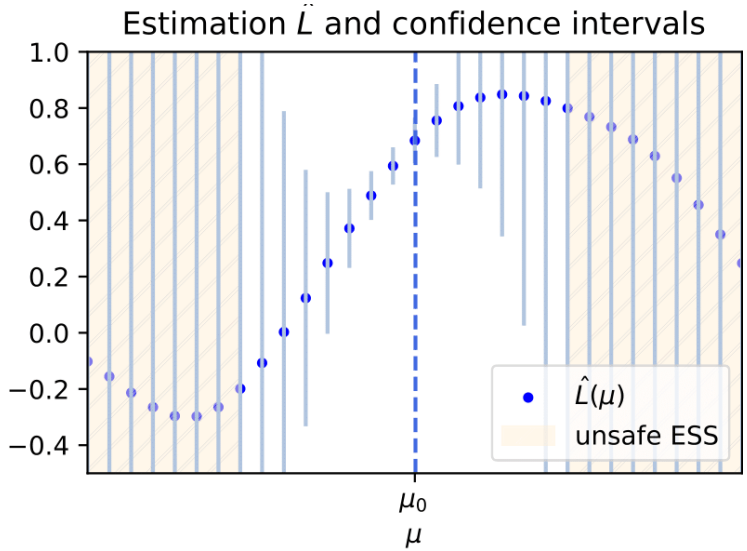
When the logging policy underexplored the action space, it is **hard** to assess with **statistical significance** the quality of policies.

The underexploration problem (Zenati et al., 2023)

When the logging policy underexplored the action space, it is **hard** to assess with **statistical significance** the quality of policies.

Consider Gaussian parametrized policies $\pi_\mu = \mathcal{N}(\mu, \sigma^2)$. We evaluate $L(\mu)$ based on samples drawn from a logging policy associated to μ_0.

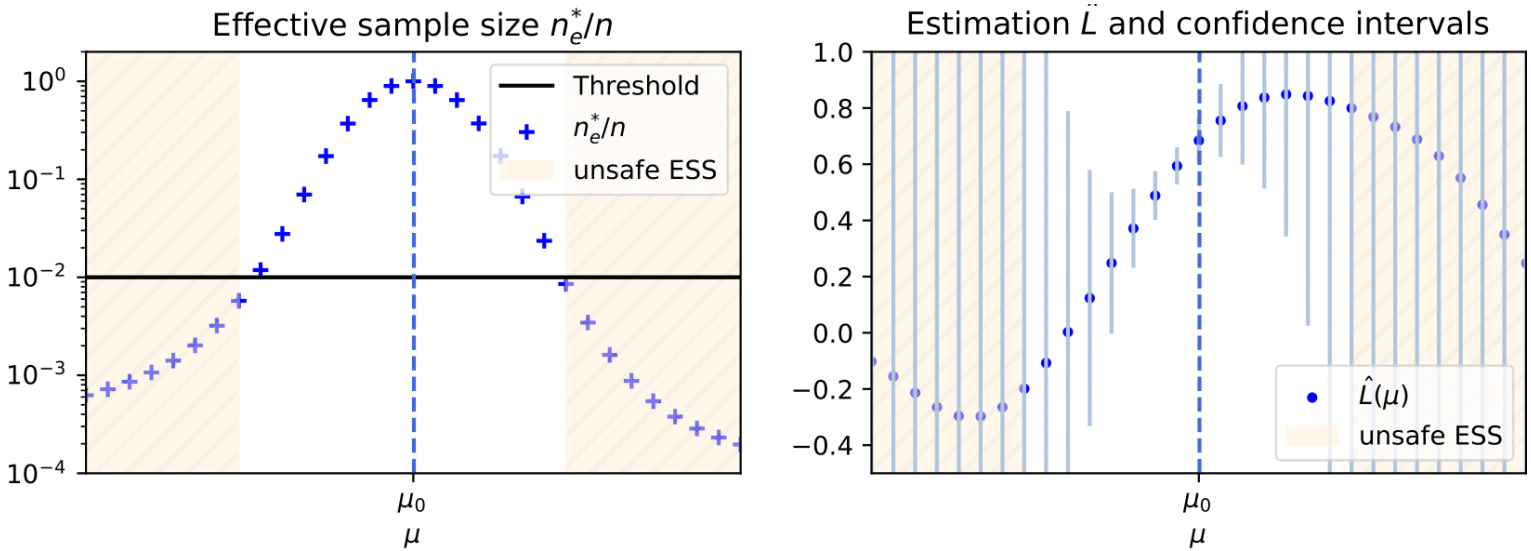




The underexploration problem (Zenati et al., 2023)

When the logging policy underexplored the action space, it is **hard** to assess with **statistical significance** the quality of policies.

Consider Gaussian parametrized policies $\pi_\mu = \mathcal{N}(\mu, \sigma^2)$. We evaluate $L(\mu)$ based on samples drawn from a logging policy associated to μ_0.

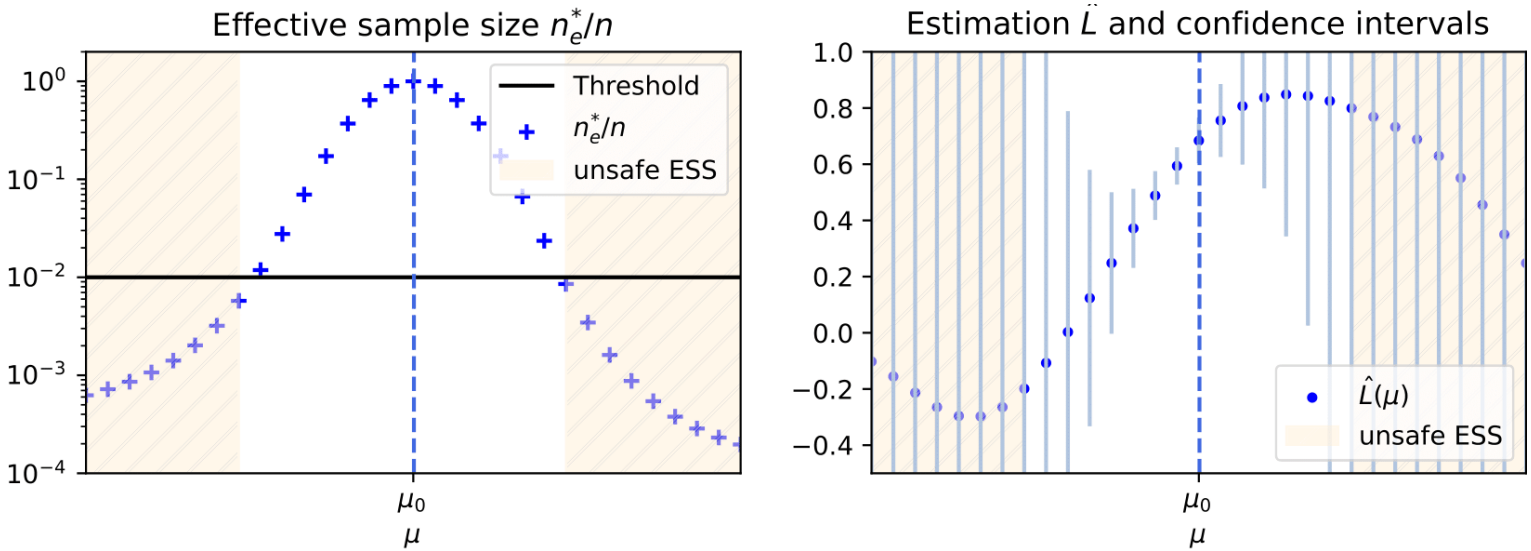


$\Longrightarrow$ effective sample sizes can be too low and confidence intervals too large.

The underexploration problem (Zenati et al., 2023)

When the logging policy underexplored the action space, it is **hard** to assess with **statistical significance** the quality of policies.

Consider Gaussian parametrized policies $\pi_\mu = \mathcal{N}(\mu, \sigma^2)$. We evaluate $L(\mu)$ based on samples drawn from a logging policy associated to μ_0.



$\Longrightarrow$ effective sample sizes can be too low and confidence intervals too large.

Therefore it is sometimes necessary to increase the sample size by collecting additional samples with **sequential** deployments.

$\Longrightarrow$ **Adaptively** collect data instead of **redeploying** the same logging model θ_0.

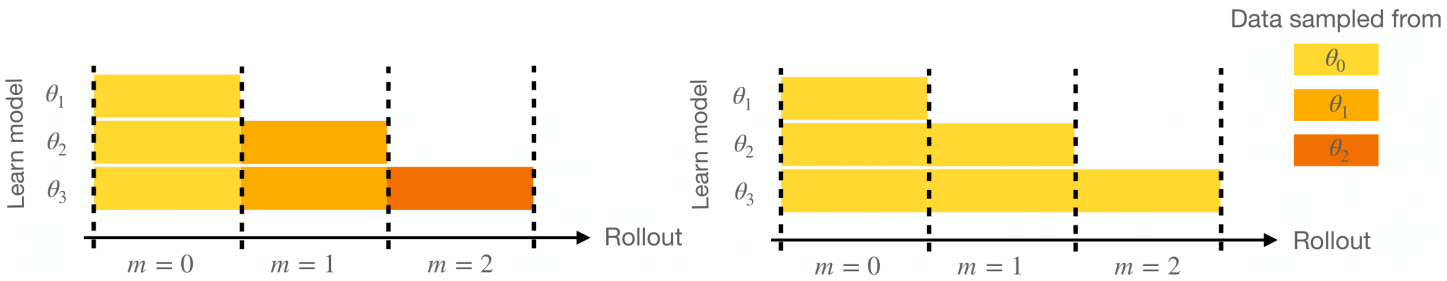


Figure: Sequential designs (left) and non-adaptive redeployment (right).

At round m, a model $\theta_m \in \Theta$ is deployed and observations $s_m = (x_{m,i}, a_{m,i}, y_{m,i})_{i=1,\dots,n_m}$ are collected to learn a new model θ_{m+1}.

Algorithm Sequential Counterfactual Risk Minimization

Input: Logged observations $(x_{0,i}, a_{0,i}, y_{0,i}, \pi_{0,i})_{i=1,\dots,n_0}$, parameter $\lambda > 0$

for $m = 0$ *to* $M - 1$ **do**

  Build $\mathcal{L}_m$ from observations s_m

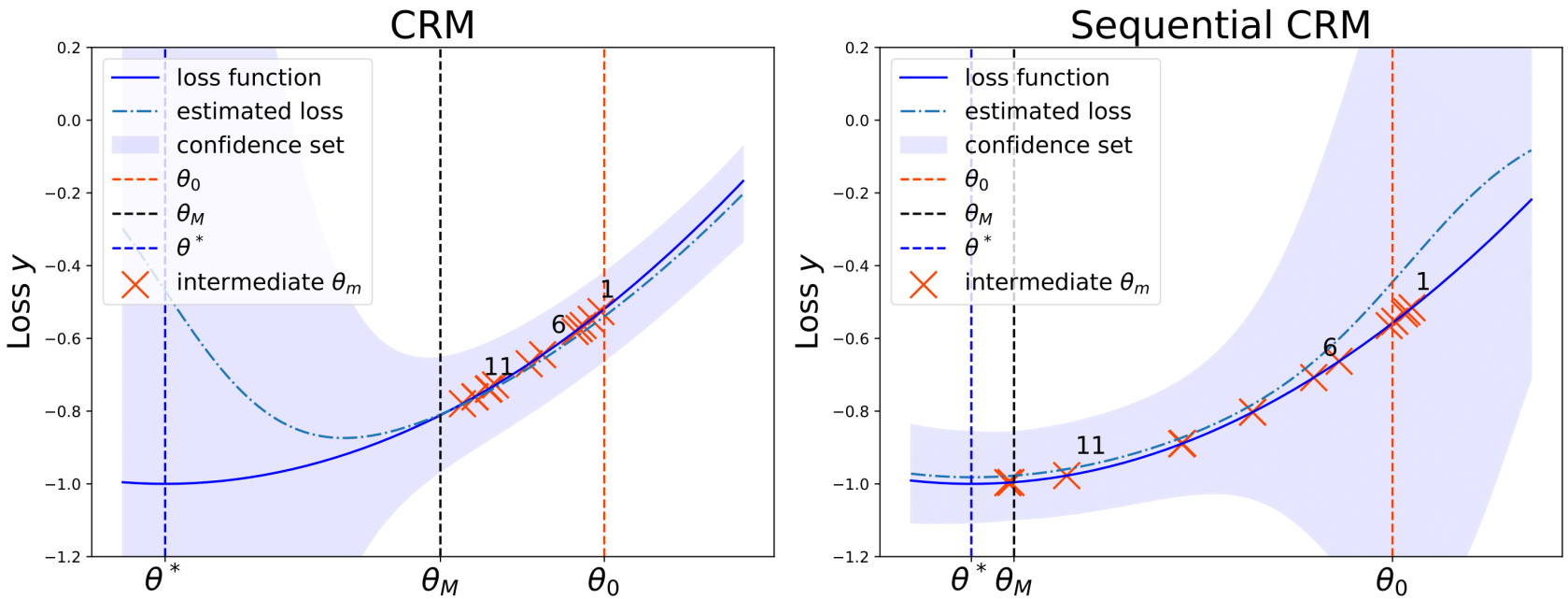
  Learn θ_{m+1}

  Deploy the model θ_{m+1} and collect observations s_{m+1}

end

Illustrating the benefits of sequential designs (Zenati et al., 2023)

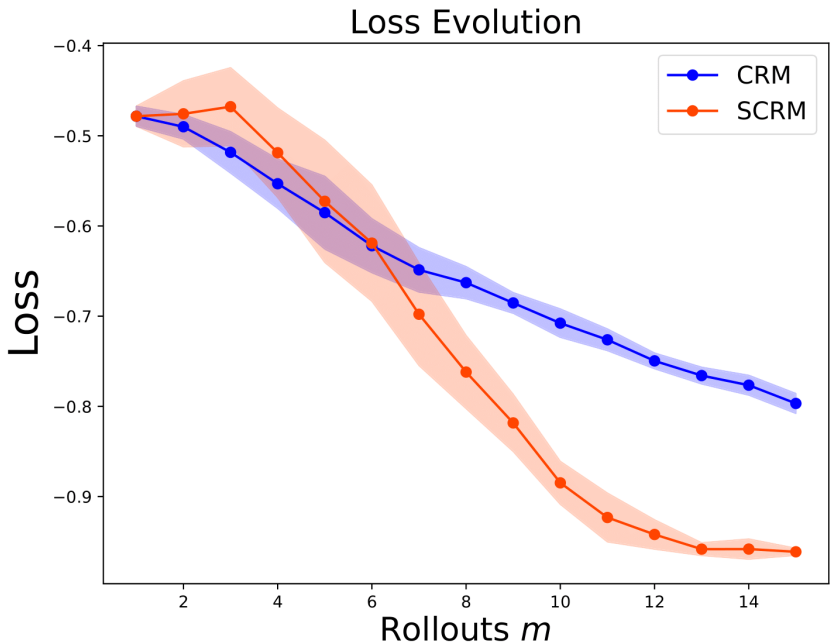
Consider Gaussian parametrized policies $\pi_\theta = \mathcal{N}(\theta, \sigma^2)$ with quadratic losses $l_t(a) = (a - y_t)^2 - 1$ where $y_t \sim \mathcal{N}(\theta^*, \sigma^2)$.



We see that the models learned with SCRM take larger optimization steps than the ones with CRM.

Illustrating the benefits of sequential designs (Zenati et al., 2023)

Consider Gaussian parametrized policies $\pi_\theta = \mathcal{N}(\theta, \sigma^2)$ with quadratic losses $l_t(a) = (a - y_t)^2 - 1$ where $y_t \sim \mathcal{N}(\theta^*, \sigma^2)$.



We see that the models learned with SCRM take larger optimization steps than the ones with CRM.

IPS-IX: Implicit exploration and controlled variance (Zenati et al., 2023)

We propose a new counterfactual estimator. Denoting $\pi_{m,i} = \pi_{\theta_m}(a_{m,i}|x_{m,i})$ and $\alpha > 0$, we change the importance weighted estimator:

$$\begin{cases} \nu_{i,m}(\theta) = y_{m,i} \min\left\{ \dfrac{\pi_{\theta,i}}{\pi_{\theta_m,i}}, \alpha \right\} & \hat{L}^{\text{cIPS}}(\theta) = \dfrac{1}{n_m} \displaystyle\sum_{i=1}^{n_m} \nu_{i,m}(\theta) \qquad \text{(cIPS)} \\ \zeta_{i,m}(\theta) = y_{m,i} \dfrac{\pi_{\theta,i}}{\pi_{\theta_m,i} + \alpha \pi_{\theta,i}} & \hat{L}^{\text{IPS-IX}}(\theta) = \dfrac{1}{n_m} \displaystyle\sum_{i=1}^{n_m} \zeta_{i,m}(\theta) \qquad \text{(IPS-IX)} \end{cases}$$

IPS-IX: Implicit exploration and controlled variance (Zenati et al., 2023)

We propose a new counterfactual estimator. Denoting $\pi_{m,i} = \pi_{\theta_m}(a_{m,i}|x_{m,i})$ and $\alpha > 0$, we change the importance weighted estimator:

$$\begin{cases} \nu_{i,m}(\theta) = y_{m,i} \min\left\{ \dfrac{\pi_{\theta,i}}{\pi_{\theta_m,i}}, \alpha \right\} & \hat{L}^{\text{cIPS}}(\theta) = \dfrac{1}{n_m} \displaystyle\sum_{i=1}^{n_m} \nu_{i,m}(\theta) \qquad \text{(cIPS)} \\ \zeta_{i,m}(\theta) = y_{m,i} \dfrac{\pi_{\theta,i}}{\pi_{\theta_m,i} + \alpha \pi_{\theta,i}} & \hat{L}^{\text{IPS-IX}}(\theta) = \dfrac{1}{n_m} \displaystyle\sum_{i=1}^{n_m} \zeta_{i,m}(\theta) \qquad \text{(IPS-IX)} \end{cases}$$

Proposition (Generalization Error Bound, informal)

Let $\theta \in \Theta$, then with high propability:

$$L(\theta) \lesssim \hat{L}_m^{\text{IPS-IX}}(\theta) + \lambda_m \sqrt{\frac{\widehat{Var}(\zeta_{i,m}(\theta) - y_{m,i})}{n_m}} + \frac{\lambda_m^2}{n_m},$$

for a λ_m depending on a complexity measure of Θ.

IPS-IX resembles the implicit exploration estimator in the EXP3-IX algorithm (Lattimore & Szepesvári, 2020).

Variance-dependent convergence guarantees (Zenati et al., 2023)

The latter motivates to solve for $m \in \{1, \dots M\}$:

$$\hat{\theta}_{m+1} \in \arg\min_{\theta \in \Theta} \hat{L}_m^{\text{IPS-IX}}(\theta) + \lambda_m \sqrt{\frac{\widehat{\text{Var}}(\zeta_{i,m}(\theta) - y_{m,i})}{n_m}}. \tag{SCRM}$$

Let $\theta^* \in \arg\min_\theta L(\theta)$.

The latter motivates to solve for $m \in \{1, \dots M\}$:

$$\hat{\theta}_{m+1} \in \arg\min_{\theta \in \Theta} \hat{L}_m^{\text{IPS-IX}}(\theta) + \lambda_m \sqrt{\frac{\widehat{\text{Var}}(\zeta_{i,m}(\theta) - y_{m,i})}{n_m}}. \qquad \text{(SCRM)}$$

Let $\theta^* \in \arg\min_\theta L(\theta)$.

Proposition (Excess Risk Bound, informal)

With high probability:

$$L(\theta_{m+1}) - L(\theta^*) \lesssim \sqrt{\nu_m^2 \frac{\log n_m}{n_m}},$$

where $\nu_m^2 = \text{Var}_{x,\theta_m}\left(\frac{\pi_{\theta^*}(a|x)}{\pi_{\theta_m}(a|x)}\right)$.

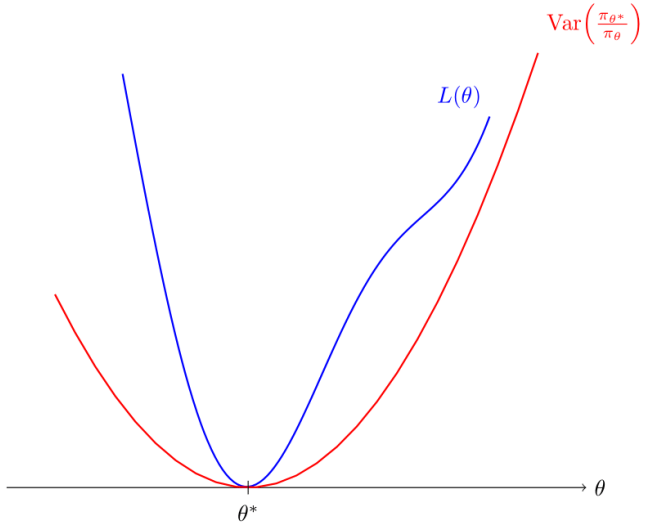
$\implies$ IPS-IX improves the dependency in ν_m^2, compared to $\nu_m^2 + 1$ in previous work (Zenati et al., 2020).

$\implies$ crucial for using these error bounds **sequentially** since $\nu_m \to 0$ if $\theta_m \to \theta^*$.

Assumption (Hölderian Error Bound)

There exist $\gamma > 0$ *and* $\beta > 0$ *such that for any* $\theta \in \Theta$,

$$\gamma \mathrm{Var}_{x,\theta}\left(\frac{\pi_{\theta^*}(x|a)}{\pi_\theta(x|a)}\right) \leq (L(\theta) - L(\theta^*))^\beta .$$



Classical bounds are of the form $\gamma d(\theta, \theta^*) \leq (L(\theta) - L(\theta^*))^\beta$ with d a distance.

Proposition (SCRM Excess Risk, informal)

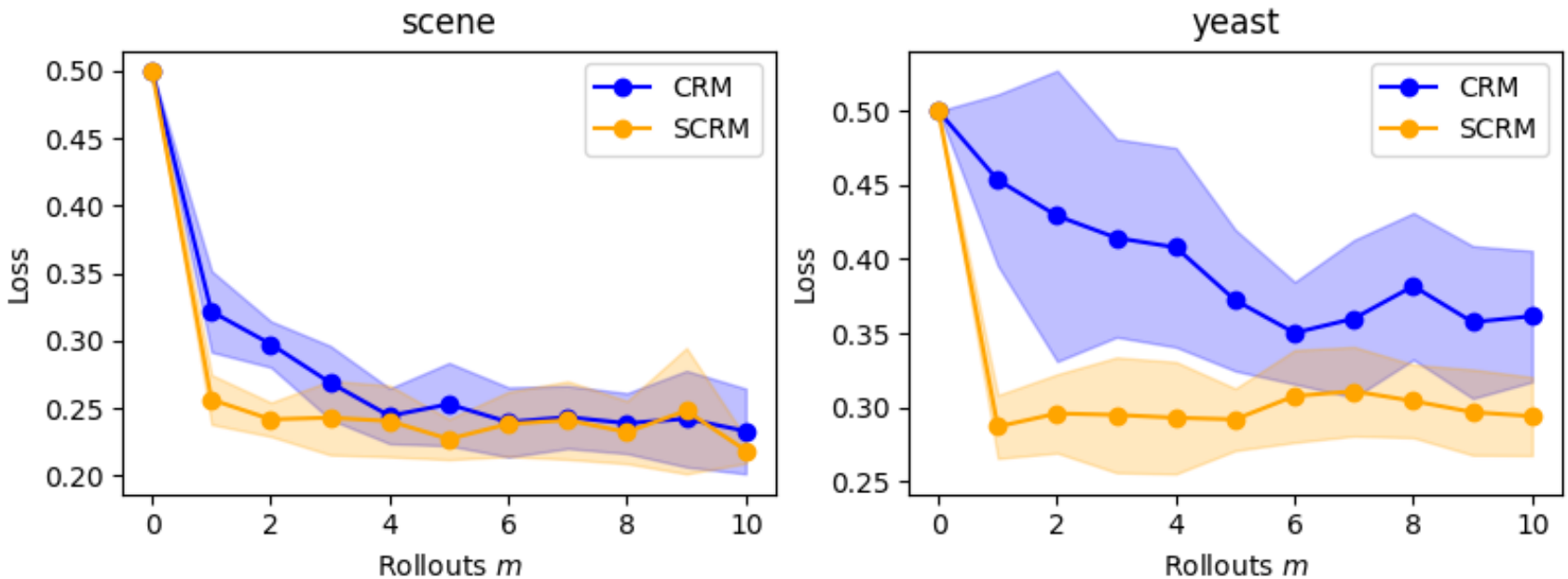
Let $n_0, n \geq 2$, **geometric** *sample sizes* $n_m = n_0 2^m$ *for* $m = 0, \ldots, M \approx \left\lfloor \log \frac{n}{n_0} \right\rfloor$.

$$L(\theta_M) - L(\theta^*) \leq O\left(n^{-\frac{1}{2-\beta}} \log n\right).$$

Fast rates on par with ERM (Bartlett & Mendelson, 2006; Van Erven et al., 2015) and when $\beta = 0$, we recover the CRM rates (Zenati et al., 2020).

SCRM in practice (Zenati et al., 2023)

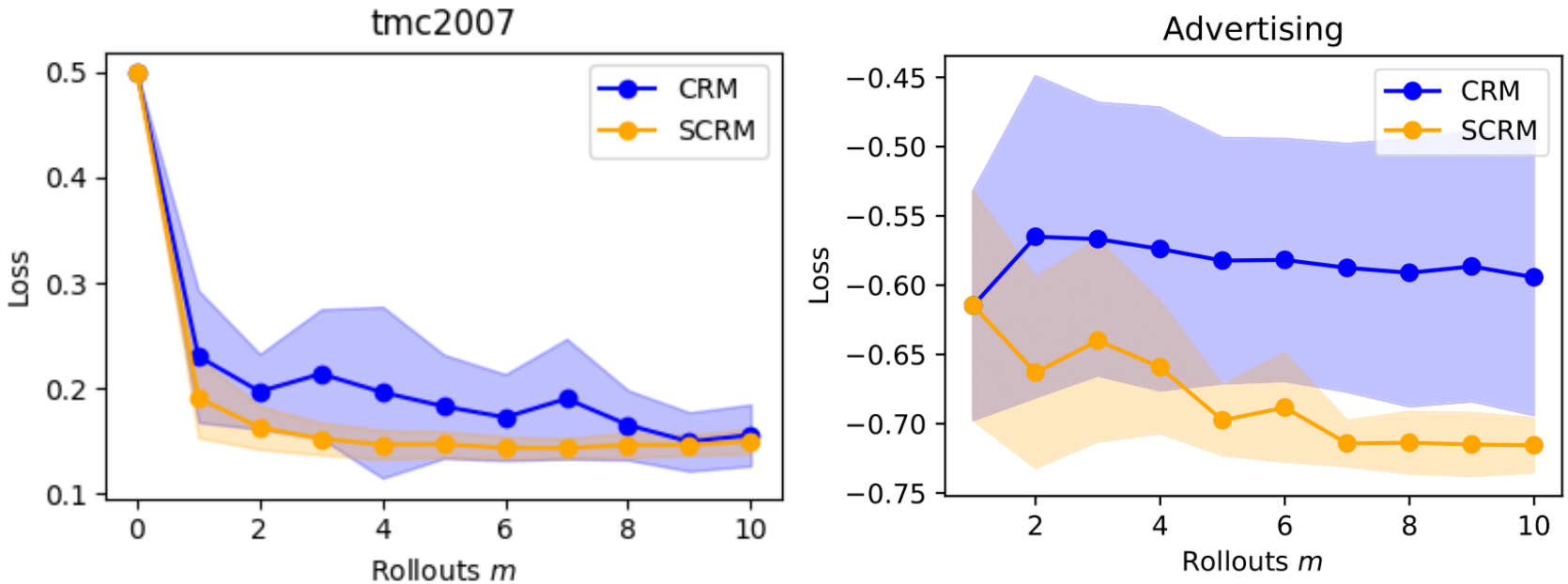
We consider discrete action real-world datasets (*Scene*, *Yeast*, *TMC2007*) and a continuous action setting (*Advertising*).



SCRM converges very fast, often within the first rollouts. Conversely, CRM needs more samples and the variance is higher.

SCRM in practice (Zenati et al., 2023)

We consider discrete action real-world datasets (*Scene*, *Yeast*, *TMC2007*) and a continuous action setting (*Advertising*).



SCRM converges very fast, often within the first rollouts. Conversely, CRM needs more samples and the variance is higher.

Perspectives

- Counterfactual policy learning in real-world applications requires an offline evaluation protocol.
- When the logging policy underexplores the action space, counterfactual learning is hard, instead **sequential** deployments (when possible) can drastically improve the learning.
- We note that the learning objective was **conservative** and it may not encourage **exploration**.

Perspectives

- Counterfactual policy learning in real-world applications requires an offline evaluation protocol.
- When the logging policy underexplores the action space, counterfactual learning is hard, instead **sequential** deployments (when possible) can drastically improve the learning.
- We note that the learning objective was **conservative** and it may not encourage **exploration**.

In some applications however, **exploration** is not that costly and online learning is possible.

Perspectives

- Counterfactual policy learning in real-world applications requires an offline evaluation protocol.
- When the logging policy underexplores the action space, counterfactual learning is hard, instead **sequential** deployments (when possible) can drastically improve the learning.
- We note that the learning objective was **conservative** and it may not encourage **exploration**.

In some applications however, **exploration** is not that costly and online learning is possible.

Examples at Criteo: recommendation, ressource allocation, hyperparameter search.

We list the following future research directions:

Counterfactual learning methods

- Investigate the use of doubly robust estimators and other control variate estimators for continuous action methods.
- In SCRM, develop guarantees to use all past samples (particularly effective in practice). To avoid introducing dependencies in the excess risk bounds, we used geometric sample sizes to discard previous samples.

Perspectives and research directions

We list the following future research directions:

Counterfactual learning methods

- Investigate the use of doubly robust estimators and other control variate estimators for continuous action methods.
- In SCRM, develop guarantees to use all past samples (particularly effective in practice). To avoid introducing dependencies in the excess risk bounds, we used geometric sample sizes to discard previous samples.

Contextual bandits

- In the stochastic setting, make a large benchmark of elimination algorithms to see their practical performance, then investigate the use of efficient methods.
- In the adversarial setting with nesting, extend to contextual information with less assumptions on feedbacks.

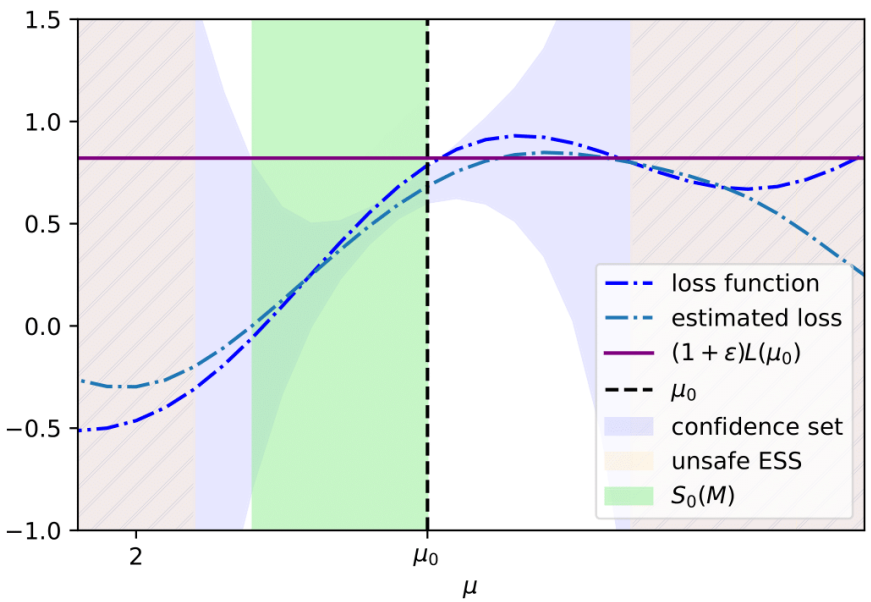
Promising research direction (i): safe SCRM

Regarding the exploration-exploitation tradeoff, investigate the optimism in the face of uncertainty (OFUL) principle in SCRM along with safety constraints:

$$\text{LCB}(\theta) = \hat{L}(\theta) + \lambda\sqrt{\frac{\hat{V}(\theta)}{n}} + \frac{\lambda^2}{(n-1)}.$$

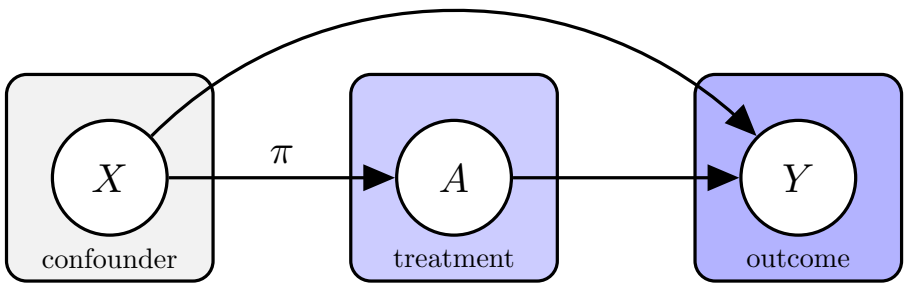
We could define a safety set of $\theta \in \Theta$ where the CRM solution lies:

$$S_0(\Theta) = \left\{\theta \in M, \text{ s.t } LCB(\theta) \leq (1+\epsilon)\hat{L}(\theta_0)\right\}.$$



Promising research direction (ii): Confounded CRM

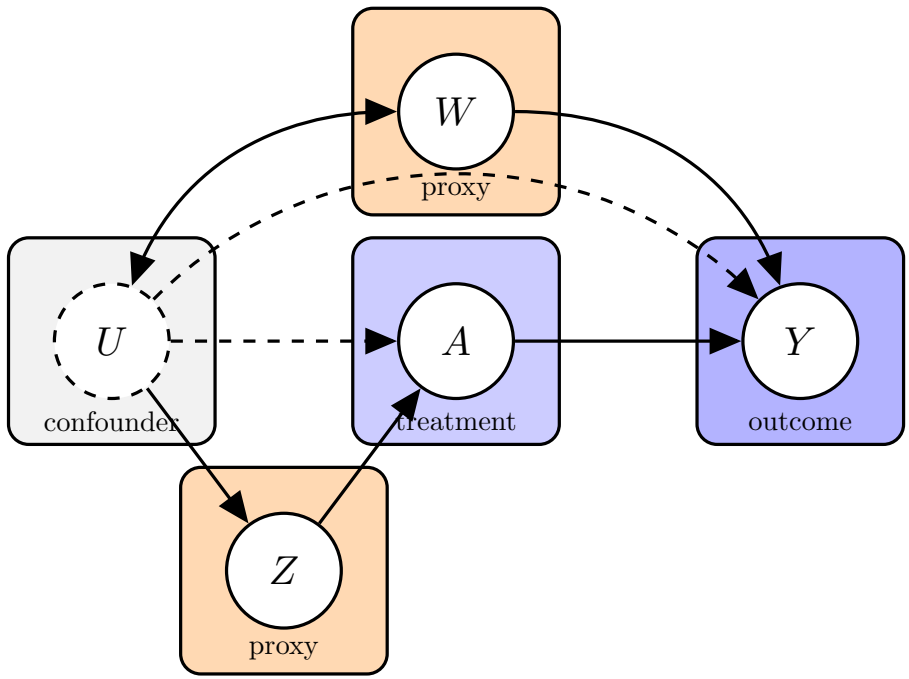
We assumed that there are no unobserved confounders to the context covariates:



However, what if this does not hold? This assumption can prove overly restrictive.

Promising research direction (ii): Confounded CRM

If we know *proxy* variables, we can use proxy causal learning (PCL) to recevor the true value functions (i.e treatment effects) (Deaner, 2018; Singh, 2020).



Employ such methods to perform confounded policy evaluation and learning!

Take home message

- Online learning is **well studied**, there are **effective** methods but it can be **risky** for some applications.

Take home message

- Online learning is **well studied**, there are **effective** methods but it can be **risky** for some applications.
- Offline counterfactual learning strategies can be preferred in that case but it is **hard** to do especially when the logging system **underexplored** the action space.

Take home message

- Online learning is **well studied**, there are **effective** methods but it can be **risky** for some applications.
- Offline counterfactual learning strategies can be preferred in that case but it is **hard** to do especially when the logging system **underexplored** the action space.
- Sequential counterfactual learning can arguably bridge the gap and take the **best of both worlds**.

Acknowledgment

- Supervisors: Eustache Diemert, Pierre Gaillard, Julien Mairal
- Co-authors: Alberto Bietti, Matthieu Martin, Panagiotis Mertikopoulos Thibaud Rahier
- My colleagues at Criteo and INRIA
- PhD jury members: Massih-Reza Amini, Olivier Cappé, Nicolò Cesa-Bianchi, Claire Vernade

References I


Abbasi-yadkori, Y., Pál, D., & Szepesvári, C. (2011).Improved algorithms for linear stochastic bandits. *Advances in Neural Information Processing Systems (NIPS)*.


Bartlett, P. L., & Mendelson, S. (2006).Empirical minimization. *Probability Theory and Related Fields*.


Calandriello, D., Carratino, L., Lazaric, A., Valko, M., & Rosasco, L. (2019).Gaussian process optimization with adaptive sketching: Scalable and no regret. *Conference on Learning Theory (COLT)*.


Calandriello, D., Carratino, L., Lazaric, A., Valko, M., & Rosasco, L. (2020).Near-linear time Gaussian process optimization with adaptive batching and resparsification. *International Conference on Machine Learning (ICML)*.


Calandriello, D., Lazaric, A., & Valko, M. (2017).Efficient second-order online kernel learning with adaptive embedding. *Adv. Neural Information Processing Systems (NIPS)*.


Chen, G., Zeng, D., & Kosorok, M. R. (2016).Personalized dose finding using outcome weighted learning. *Journal of the American Statistical Association*, *111*(516), 1509–1521.


Deaner, B. (2018). Proxy controls and panel data.

References II

Dudik, M., Erhan, D., Langford, J., & Li, L. (2014).Doubly robust policy evaluation and optimization. *Statist. Sci.*, *29*(4), 485–511. https://doi.org/10.1214/14-STS500

Dylan J. Foster, A. R. (2020).Beyond ucb: Optimal and efficient contextual bandits with regression oracles.. *ICML*.

Horvitz, D. G., & Thompson, D. J. (1952).A generalization of sampling without replacement from a finite universe. *Journal of the American Statistical Association*, *47*(260), 663–685.

Kallus, N., & Zhou, A. (2018).Policy evaluation and optimization with continuous treatments. *International Conference on Artificial Intelligence and Statistics (AISTATS)*.

Krause, A., & Ong, C. S. (2011).Contextual gaussian process bandit optimization. *Adv. Neural Information Processing Systems (NIPS)*.

Lattimore, T., & Szepesvári, C. (2020). *Bandit algorithms*. Cambridge University Press.

Littlestone, N., & Warmuth, M. K. (1994).The weighted majority algorithm. *Information and Computation*, *108*(2), 212–261.

Martin, M., Mertikopoulos, P., Rahier, T., & Zenati, H. (2022).Nested bandits. *International Conference on Machine Learning (ICML)*.

References III

- Naoki Abe, P. M. L. (1999).Associative reinforcement learning using linear probabilistic concepts. *ICML*.
- Owen, A. B. (2013). *Monte carlo theory, methods and examples*.
- Singh, R. (2020).Kernel methods for unobserved confounding: Negative controls, proxies, and instruments.
- Swaminathan, A., & Joachims, T. (2015).Counterfactual risk minimization: Learning from logged bandit feedback. *ICML*.
- Valko, M., Korda, N., Munos, R., Flaounas, I., & Cristianini, N. (2013).Finite-time analysis of kernelised contextual bandits. *Conference on Uncertainty in Artificial Intelligence (UAI)*.
- Van Erven, T., Grünwald, P. D., Mehta, N. A., Reid, M. D., & Williamson, R. C. (2015).Fast rates in statistical and online learning. *Journal of Machine Learning Research*.
- Zenati, H., Bietti, A., Diemert, E., Mairal, J., Martin, M., & Gaillard, P. (2022).Efficient kernelized UCB for contextual bandits. *International Conference on Artificial Intelligence and Statistics (AISTATS)*.

References IV


Zenati, H., Bietti, A., Martin, M., Diemert, E., Gaillard, P., & Mairal, J. (2020).Counterfactual learning of stochastic policies with continuous actions: From models to offline evaluation. https://arxiv.org/abs/2004.11722


Zenati, H., Bietti, A., Martin, M., Diemert, E., & Mairal, J. (2020).Optimization approaches for counterfactual risk minimization with continuous actions. *International Conference on Learning Representation (ICLR), Causal Learning for Decision Making Workshop*.


Zenati, H., Diemert, E., Martin, M., Mairal, J., & Gaillard, P. (2023).Sequential counterfactual risk minimization. *International Conference on Machine Learning (ICML)*.